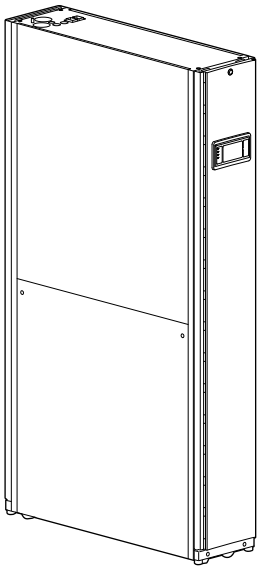
Uniflair™ Chilled-Water InRow Cooling

ACRC300 Series

Installation Manual

990-4705D-001
Release Date:01/2026



www.se.com

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This product is for industrial use only. It should only be used for the functions for which it has been designed as set out in this manual. User must evaluate and take adequate precautions to address risks associated with use of this product in environments and/or processes not specifically addressed in this manual (e.g. heavy industry, medical, marine environments, railway, etc.).

This product should be installed, configured, used, serviced, maintained, replaced or have similar work carried out on it only by suitably qualified, trained, experienced and competent personnel who hold any necessary authorizations (e.g. licenses, permits or certifications) to perform such work. User must ensure that all work is carried out in compliance with the manufacturer’s instructions (including the product labels/ markings, technical specification manual, installation manual, and operation and maintenance manual) and with all applicable laws, regulations, standards and guidance (including standards and guidance applicable to the installation location).

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Safety

Important Safety Instructions — SAVE THESE INSTRUCTIONS

Read these instructions carefully and look at the equipment to become familiar with it before trying to install, operate, service or maintain it. The following safety messages may appear throughout this manual or on the equipment to warn of potential hazards or to call attention to information that clarifies or simplifies a procedure.



The addition of this symbol to a “Danger” or “Warning” safety message indicates that an electrical hazard exists which will result in personal injury if the instructions are not followed.



This is the safety alert symbol. It is used to alert the user to potential personal injury hazards. Obey all safety messages with this symbol to avoid possible injury or death.



This is the service indicator symbol. It is used to advise the user to read the technical manual.



This is the operator’s manual symbol. It is used to advise the user to read the operating instructions.

⚠DANGER

DANGER indicates a hazardous situation which, if not avoided, **will result in** death or serious injury.

Failure to follow these instructions will result in death or serious injury.

⚠WARNING

WARNING indicates a hazardous situation which, if not avoided, **could result in** death or serious injury.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠CAUTION

CAUTION indicates a hazardous situation which, if not avoided, **could result in** minor or moderate injury.

Failure to follow these instructions can result in injury or equipment damage.

NOTICE

NOTICE is used to address practices not related to physical injury. The safety alert symbol shall not be used with this type of safety message.

Failure to follow these instructions can result in equipment damage.

Please Note

Electrical equipment should only be installed, operated, serviced, and maintained by qualified personnel. No responsibility is assumed by Schneider Electric for any consequences arising out of the use of this material.

A qualified person is one who has skills and knowledge related to the construction, installation, and operation of electrical equipment and has received safety training to recognize and avoid the hazards involved.

Always abide strictly by local laws and regulations in the place of installation.

Safety During Installation

⚠⚠DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

- Apply appropriate personal protective equipment (PPE) and follow safe electrical work practices.
- This equipment must be installed and serviced by qualified and trained personnel only.
- Turn off all power supplying this equipment before working on or inside the equipment.
- Replace all devices, doors, and covers before turning on power to this equipment.

Failure to follow these instructions will result in death or serious injury.

⚠WARNING

HAZARD TO EQUIPMENT OR PERSONNEL

This equipment is not to be operated or installed by persons with reduced physical, sensory, or mental capabilities, or persons lacking experience or knowledge. Children are not to operate or play on or around this equipment.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠WARNING

HAZARD OF EQUIPMENT FALLING OVER

- Use two or more persons at all times to move or turn this equipment.
- Always push, pull, or turn while facing the front and rear of this equipment. Never push, pull, or turn while facing the sides of this equipment.
- Slowly move this equipment across uneven surfaces or door thresholds.
- Lower leveling feet to floor when this equipment is at rest.
- Lower leveling feet and attach joining brackets to adjacent racks when this equipment is in final position.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠WARNING

HAZARD FROM MOVING PARTS

Keep hands, clothing, and jewelry away from moving parts. Check the equipment for foreign objects before closing the doors and starting the equipment.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠WARNING

HAZARD TO EQUIPMENT OR PERSONNEL

All work must be performed by Schneider Electric qualified and trained personnel.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

NOTICE

RISK FROM SHARP EDGES

- Edges of the knock-out panel may be sharp.
- Use a grommet, bushing, or other protective covering for the section of power cable routed through the knock-out panel.

Failure to follow these instructions can result in equipment damage.

General Information

Translation Reference

The original version of this manual was written in English.

Manuals written in other languages have been translated from the original English version.

Document Overview

Original Instructions

These are the original instructions provided by the manufacturer.

Save These Instructions

This manual contains important instructions that must be followed during the installation of this equipment.

Manual Updates

Schneider Electric™ policy is one of continuous technological innovation and the company reserves the right to amend any data herein without prior notice. The images shown in this manual are for descriptive purposes only.

NOTE: Unit images and component identification information are examples only.

For any updates to this manual, please contact Schneider Electric™ providing the related part number displayed on the manual back cover.

Cross-Reference Symbol Used in This Manual



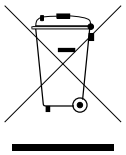
See another section of this document or another document for more information about this subject.

Abbreviations

The following are abbreviations and terms used in this manual:

- EEV: Electronic expansion valve
- VFD/VSD: Variable-frequency drive/variable-speed drive
- CW: Chilled Water
- BMS: Building management system
- ATS: Automatic transfer switch
- HACS: Hot aisle containment system
- CACS: Cold aisle containment system
- RACS: Rack aisle containment system

Waste Electrical and Electronic Equipment (WEEE) Disposal



Schneider Electric products comply with international directives on the Restriction of Hazardous Substances (RoHS) in electronic and electrical equipment and the disposal of Waste Electrical and Electronic Equipment (WEEE). Dispose of any waste electronic or electrical equipment with the appropriate recycling center. Contact Schneider Electric for assistance.

At the end of an EEE (Electrical and Electronic Equipment) useful life, any battery included in the same must be removed and separated according to the instructions provided by the supplier, before disposing of the product. Used batteries must be disposed of at an appropriate waste collection center, as required by local regulations.

Environmental Considerations

Wind

The equipment is not intended for installation in areas of high wind. Consult the sales representative for information on any applicable options for installation in areas of high wind.

Earthquakes

The equipment is not intended for installation in areas at risk of seismic activity. Consult the sales representative for information on any applicable options for installation in areas at risk of seismic activity.

ATEX

The equipment is not intended for use in potentially explosive atmospheres and does not comply with Directive 2014/34 / EU (ATEX).

Dust

The equipment is not intended for use in dusty environments and in environments with conductive dust.

Corrosion

The equipment is not intended for use in a potentially corrosive environment.

Regulation on Fluorinated Greenhouse Gases

According to Regulation (EU) No. 2024/573 on fluorinated greenhouse gases, operators of stationary equipment that contain fluorinated greenhouse gases shall ensure that the recovery of those gases is carried out by natural persons that hold

the relevant certificates provided for by Article 10, so that those gases are recycled, reclaimed or destroyed.

NOTE: 'Recovery' means the collection and storage of fluorinated greenhouse gases from products, including containers, and equipment during maintenance or servicing or prior to the disposal of the products or equipment.

IMPORTANT: If Regulation (EU) No. 2024/573 is not in force in the country of installation, fluorinated greenhouse gases must be disposed of as required by local and national codes and regulations.

Radio Frequency Interference

NOTE: The Radio Frequency Interference directive applies to units installed in US and in Canada only.

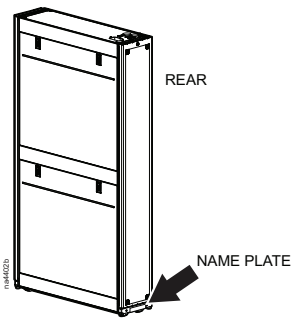
Cooling units comply with part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. These devices may not cause harmful interference.
2. These devices must accept any interference received, including interference that may cause undesired operation.

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Unit Overview

Model Identification



The model number can be found on the outside of the shipping crate and on the name plate located on the unit as shown. Use the table below to verify that the equipment is the right type and voltage.

SKU	Width	Range of Capacity	Voltage	Phase	Frequency	Power Connection
ACRC301S	300 mm	Up to 45 kW	100–240V	1	50/60 Hz	NEMA L5-20P/ IEC 309-16A
ACRC301H	300 mm	Up to 60 kW	208–230V	1	50/60 Hz	HARDWIRED

Receiving and Inspecting the Equipment

Uniflair Chilled-Water InRow air conditioner has been tested and inspected for quality assurance before shipment from Schneider Electric. Carefully inspect both the exterior and interior of the equipment immediately upon receipt to ensure that the equipment has not been damaged during transit.

Verify that all parts ordered were received as specified and that the equipment is the correct type, size, and voltage.

Storing the Cooling Unit Before Installation

NOTICE
DAMAGE FROM EXPOSURE Leaving the equipment uncovered and exposed to possible damage from the environment will void the factory warranty. **Failure to follow these instructions can result in equipment damage.**

If storing the unit for a period of time, the following procedures must be followed:

- The storage area must be dry (<85% R.H.) and protected against minimum storage temperatures –23°C (-9.4°F) and extreme storage temperatures 50° C (120°F).
- The unit should remain in its original packaging if stored for long periods of time.
- Max altitude: 2,000 m (6,562 ft)

Filing a Claim

If damage is identified on receipt of the equipment, note the damage on the bill of lading and file a damage claim with the shipping company. Contact Schneider Electric Worldwide Customer Support at one of the numbers listed on the Web page reported on the back page of this manual for information about how to file a claim with the shipping company*. The shipping claim must be filed at the receiving end of the delivery and shall be sent no later than 30 days upon unit receipt.

NOTE: In case of shipping damage, do not operate the equipment. Keep all packaging for inspection by the shipping company and contact Schneider Electric.

*e.g. a freight forwarder or an insurance company, depending on shipment arrangement

Storing the Cooling Unit Before Installation

NOTICE
DAMAGE FROM EXPOSURE Leaving the equipment uncovered and exposed to possible damage from the environment will void the factory warranty. **Failure to follow these instructions can result in equipment damage.**

If storing the unit for a period of time, the following procedures must be followed:

- The storage area must be dry (<85% R.H.) and protected against minimum storage temperatures –23°C and extreme storage temperatures 50°C.
- The unit should remain in its original packaging if stored for long periods of time.

Moving the Equipment

⚠WARNING

HAZARD OF EQUIPMENT FALLING OVER

- Use two or more persons at all times to move or turn this equipment.
- Always push, pull, or turn while facing the front and rear of this equipment. Never push, pull, or turn while facing the sides of this equipment.
- Slowly move this equipment across uneven surfaces or door thresholds.
- Lower leveling feet to floor when this equipment is at rest.
- Lower leveling feet and attach joining brackets to adjacent racks when this equipment is in final position.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

The recommended tools for moving equipment while it is still on the pallet include the following:

FORKLIFT

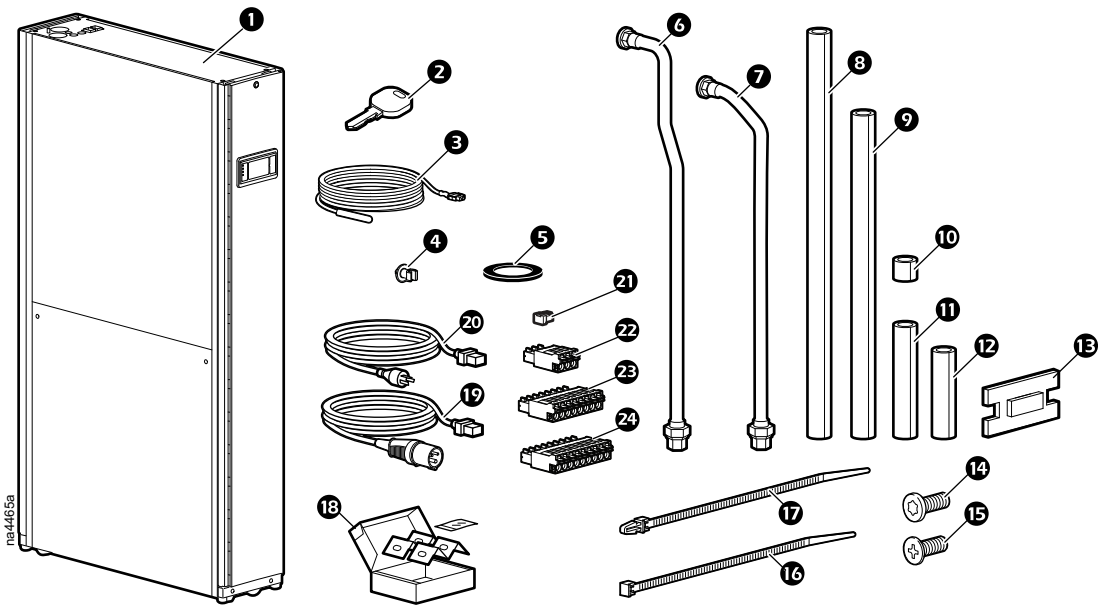


PALLET JACK



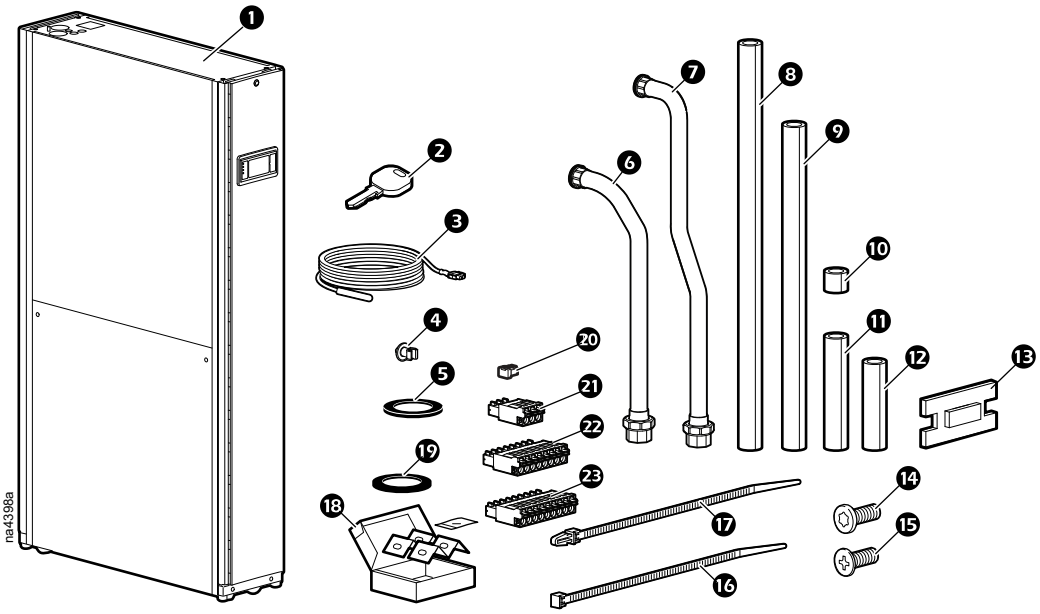
Inventory

Package Contents—ACRC301S



Item	Description	Quantity	Item	Description	Quantity
❶	InRow RC—ACRC301S	1	⓭	Insulation, bracket	2
❷	Front and rear panel key	2	⓮	TORX screw	4
❸	Remote temperature probe	1	⓯	Phillips screw, black	4
❹	Remote temperature probe clip	3	⓰	Wire tie	3
❺	Pipe gasket	4	⓱	Barb-type wire tie	3
❻	Pipe, outlet	1	⓲	Bolt down kit	1
❼	Pipe, inlet	1	⓳	Power cord, IEC 309-16A to C19	2
❽	Insulation, outlet pipe (above clamp)	1	⓴	Power cord, NEMA L5-20P to C19	2
❾	Insulation, inlet pipe (above clamp)	1	㉑	Terminator	1
❿	Insulation, union	2	㉒	Terminal plug, 3-position	2
⓫	Insulation, outlet pipe (below clamp)	1	㉓	Connector, 8-tab	1
⓬	Insulation, inlet pipe (below clamp)	1	㉔	Connector, 9-tab	1

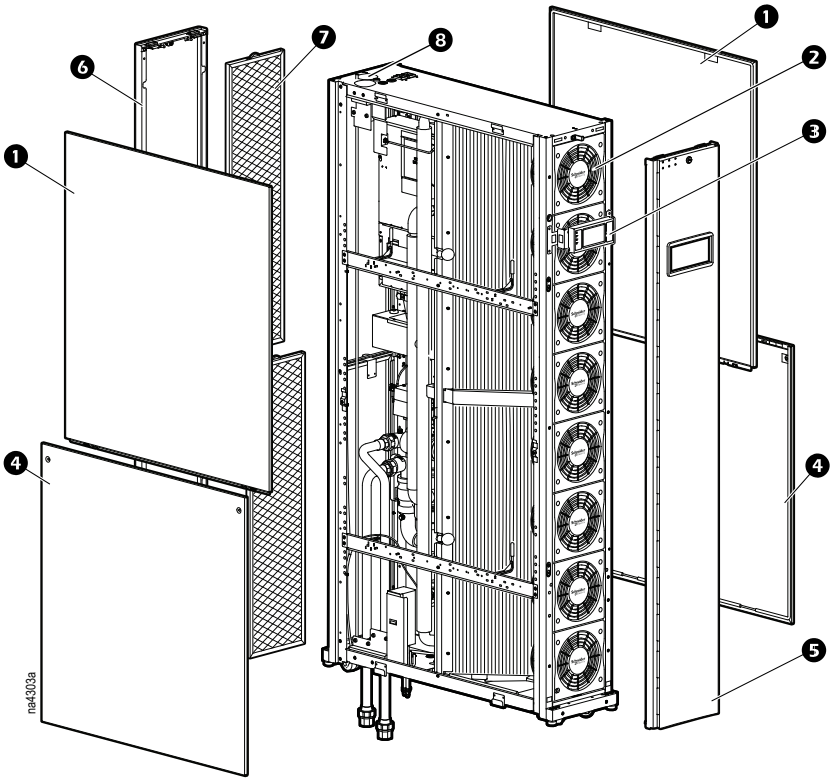
Package Contents—ACRC301H



Item	Description	Quantity	Item	Description	Quantity
❶	InRow RC—ACRC301H	1	⓭	Insulation, clamp	2
❷	Front and rear panel key	2	⓮	TORX screw	4
❸	Remote temperature probe	1	⓯	Phillips screw, black	4
❹	Remote temperature probe clip	3	⓰	Wire tie	3
❺	Pipe gasket, unit pipes	2	⓱	Barb-type wire tie	3
❻	Pipe, inlet	1	⓲	Bolt down kit	1
❼	Pipe, outlet	1	⓳	Pipe gasket, client connections	2
❽	Insulation, outlet pipe (above clamp)	1	⓴	Terminator	1
❾	Insulation, inlet pipe (above clamp)	1	㉑	Terminal plug, 3-position	2
❿	Insulation, pipe unions	2	㉒	Connector, 8-tab	1
⓫	Insulation, inlet pipe (below clamp)	1	㉓	Connector, 9-tab	1
⓬	Insulation, outlet pipe (below clamp)	1			

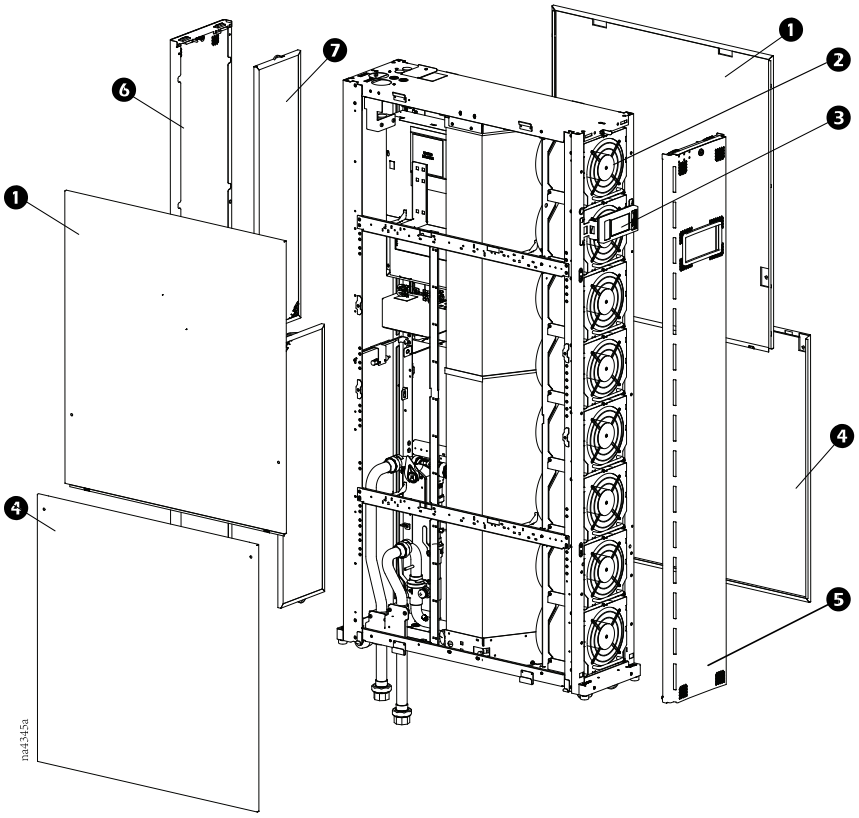
Component Identification

External Components—ACRC301S



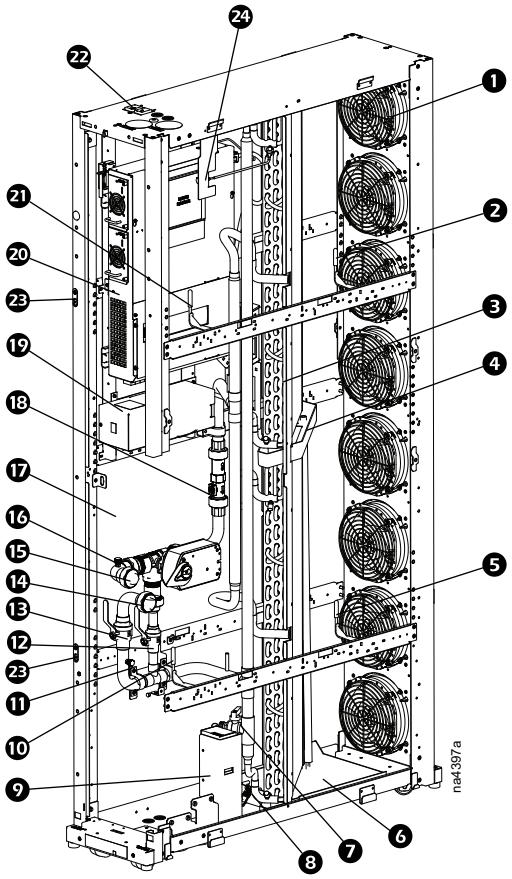
Item	Description
❶	Upper side panels
❷	Fans
❸	Display interface
❹	Lower side panels
❺	Front panel
❻	Rear panel
❼	Air filters
❽	Upper piping installation holes

External Components—ACRC301H



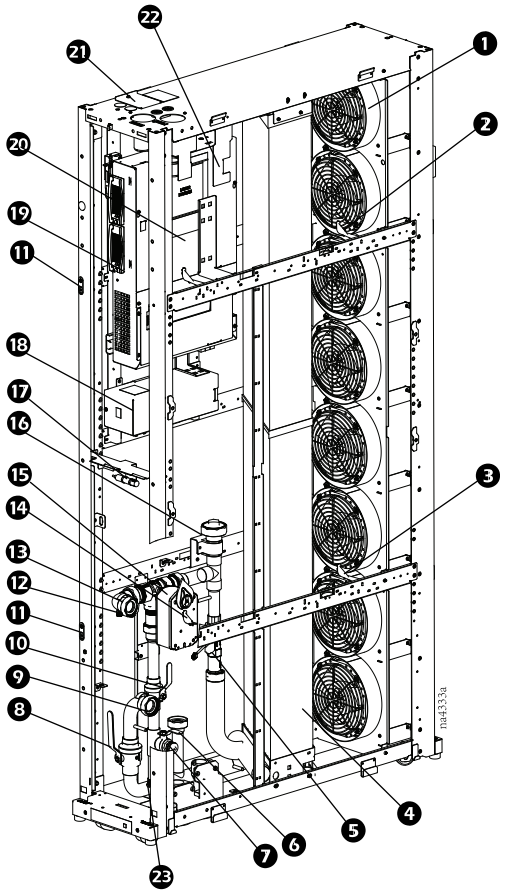
Item	Description
❶	Upper side panels
❷	Fans
❸	Display interface
❹	Lower side panels
❺	Front panel
❻	Rear panel
❼	Air filters
❽	Upper piping installation holes

Internal Components—ACRC301S



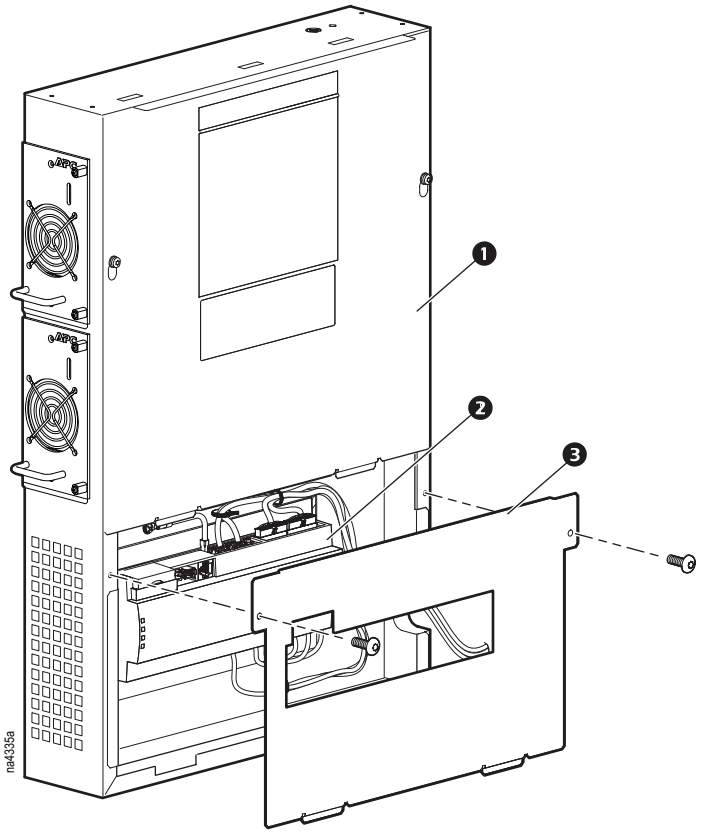
Item	Description	Item	
❶	Fan	⓭	Inlet shutoff valve (1 in.)
❷	Supply air temperature sensor, top	⓮	Pipe connection, water inlet
❸	Coil	⓯	Pipe connection, water outlet
❹	Condensate drain pan, top	⓰	Service valve (outlet water pressure test)
❺	Supply air temperature sensor, bottom	⓱	3-way actuator valve
❻	Condensate drain pan, bottom	⓲	Flow meter
❼	Drain valve	⓳	Auto-transfer switch (ATS) box
❽	Condensate drain pan float switch	⓴	Electrical box
❾	Condensate pump	㉑	Return air temperature sensors
❿	Return air temperature sensor, bottom	㉒	Power cord connections, top
⓫	Service valve (inlet water pressure test)	㉓	Rack joining brackets
⓬	2-way–(3/4-in.) bypass shut-off valve	㉔	Service valve (air purging)

Internal Components—ACRC301H



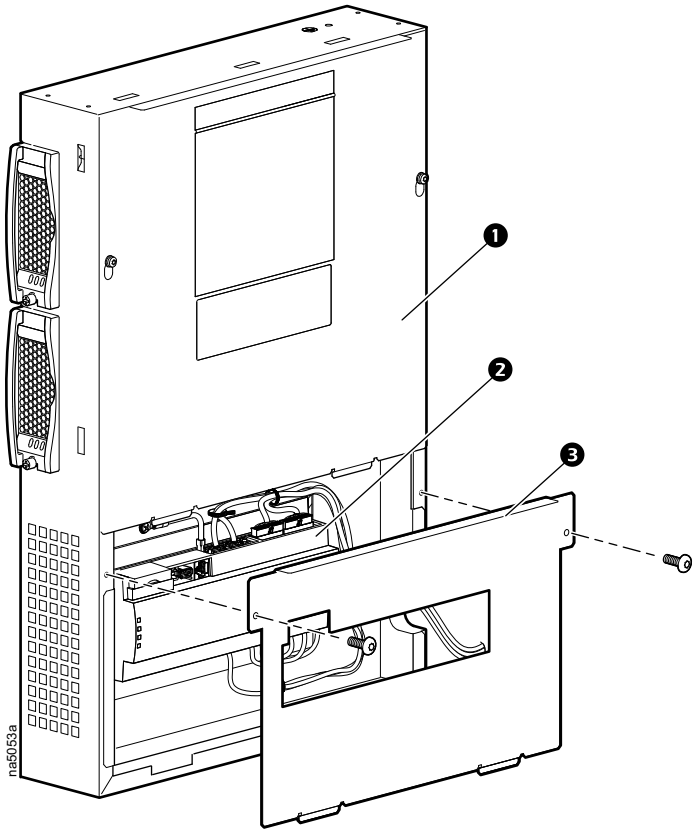
Item	Description	Item	
❶	Fan	⓭	Pipe connection, water outlet
❷	Supply air temperature sensor, top	⓮	Return air temperature sensor, bottom
❸	Supply air temperature sensor, bottom	⓯	3-way valve actuator
❹	Coil	⓰	Top pipe connection for optional recirculation pump
❺	Flow meter	⓱	Humidity sensor
❻	Bottom pipe connection for optional recirculation pump	⓲	Auto transfer switch (ATS) box
❼	Drain valve	⓳	Electrical box
❽	Inlet shutoff valve (1 1/4-in.)	⓴	Return air temperature sensor, top
❾	Pipe connection, water inlet	㉑	Junction box, top position
❿	2-way valve (1-in.) bypass shut-off	㉒	Service valve (air purging)
⓫	Rack joining brackets	㉓	Service valve (inlet water pressure test)
⓬	Service valve (outlet water pressure test)		

User interface connection panel—ACRC301S



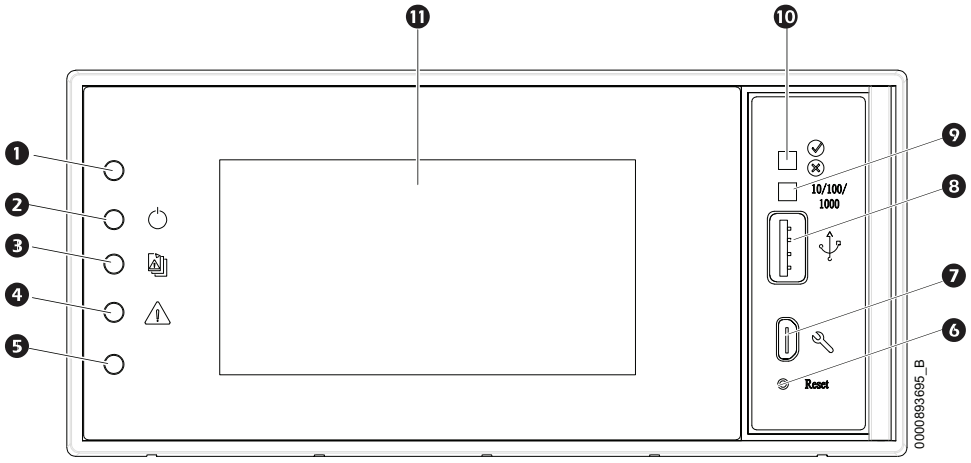
Item	Description
❶	Upper electrical box cover
❷	User interface connection panel
❸	Control module cover

User interface connection panel—ACRC301H



Item	Description
1	Upper electrical box cover
2	User interface connection panel
3	Control module cover

Display Interface



Item	Description	Function
1	Not used	
2	Power LED	The cooling unit is powered when the LED is illuminated. Unit firmware is updating when LED is blinking.

Item	Description	Function
❸	Check log LED	When this LED is illuminated, a new entry has been made to the event log.
❹	Alarm LED	Displays current alarm condition of unit.
❺	Not used	
❻	Display reset button	Resets the display microprocessor. This has no effect on the air conditioner controller.
❼	Micro USB	• Used as serial configuration port. Connect the display interface with a local computer to configure initial network settings or to access the command line interface (CLI). • Used as service port.
❽	USB-A port	Supports firmware upgrades.
❾	Link-RX/TX (10/100/1000) LED	Displays current network link status.
❿	Status LED	Displays current network management card status.
⓫	LCD display	4.3 in. touch-screen color display.

Location, Power, and Water Considerations

Room Preparation

During the design of the data center, consider ease of entry for the equipment, floor loading factors, and accessibility to piping and wiring.

Seal the room with a vapor barrier to minimize moisture infiltration. (Polyethylene film is recommended for ceiling and wall applications.) Apply rubber- or plastic-based paints to concrete walls and floors.

Insulate the room to minimize the influence of exterior heat loads. Use the minimum required amount of fresh air for make up to comply with local and national codes and regulations. Fresh air imposes extreme load variation on the cooling equipment from summer to winter and causes increased operating costs.

The cooling unit is a sensible-cooling, air conditioning unit for in-row use in data centers. The cooling unit does not have humidification or dehumidification control. Room humidity must be within acceptable operating conditions before starting the cooling unit. If operated in spaces where the humidity is in the unacceptable operating conditions section of the Operating Guidelines chart (see the InRow CW Operation and Maintenance manual), the cooling unit condenses excess water vapor from the air. For units that contain condensate pumps, this condition will exceed the pumps pumping capacity, causing the cooling unit to send an alarm and shut down to avoid overflowing the condensate pan. The condensate pump will run until the fluid level in the pan is reduced, and the alarm is automatically reset. The cooling unit self-regulates in this manner until normal operating conditions are present.

Incoming Power Supply Requirements

⚠⚠WARNING

ELECTRICAL HAZARD

- Electrical service must conform to local and national electrical codes and regulations.
- The equipment must be grounded.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

See the nameplate on the unit to determine the maximum possible current draw of the cooling unit. Provide a single circuit with sufficient capacity to handle all loads. Do not connect two InRow cooling units with the same branch circuit.

Chilled Water Temperature Requirements

For ACRC301H units, the chilled water temperature needs to be higher than the dew point of the room. If the dew point is too high, an optional dew point control pump is available for installation in ACRC301H units. The InRow CW units do not provide precision humidity control for the IT environment.

Cooling Unit Location

Refer to the floor layout drawing for the exact placement of the cooling unit in the row of IT equipment. This layout drawing can be found in the Configure-to-Order

(CTO) report or the engineering specification drawings provided by the Consulting Engineer.

Due to potentially high noise levels during peak loads, Uniflair InRow cooling units are not intended to be used in an occupied office environment. It is recommended wearing ear protection if prolonged exposure to high noise level is expected.

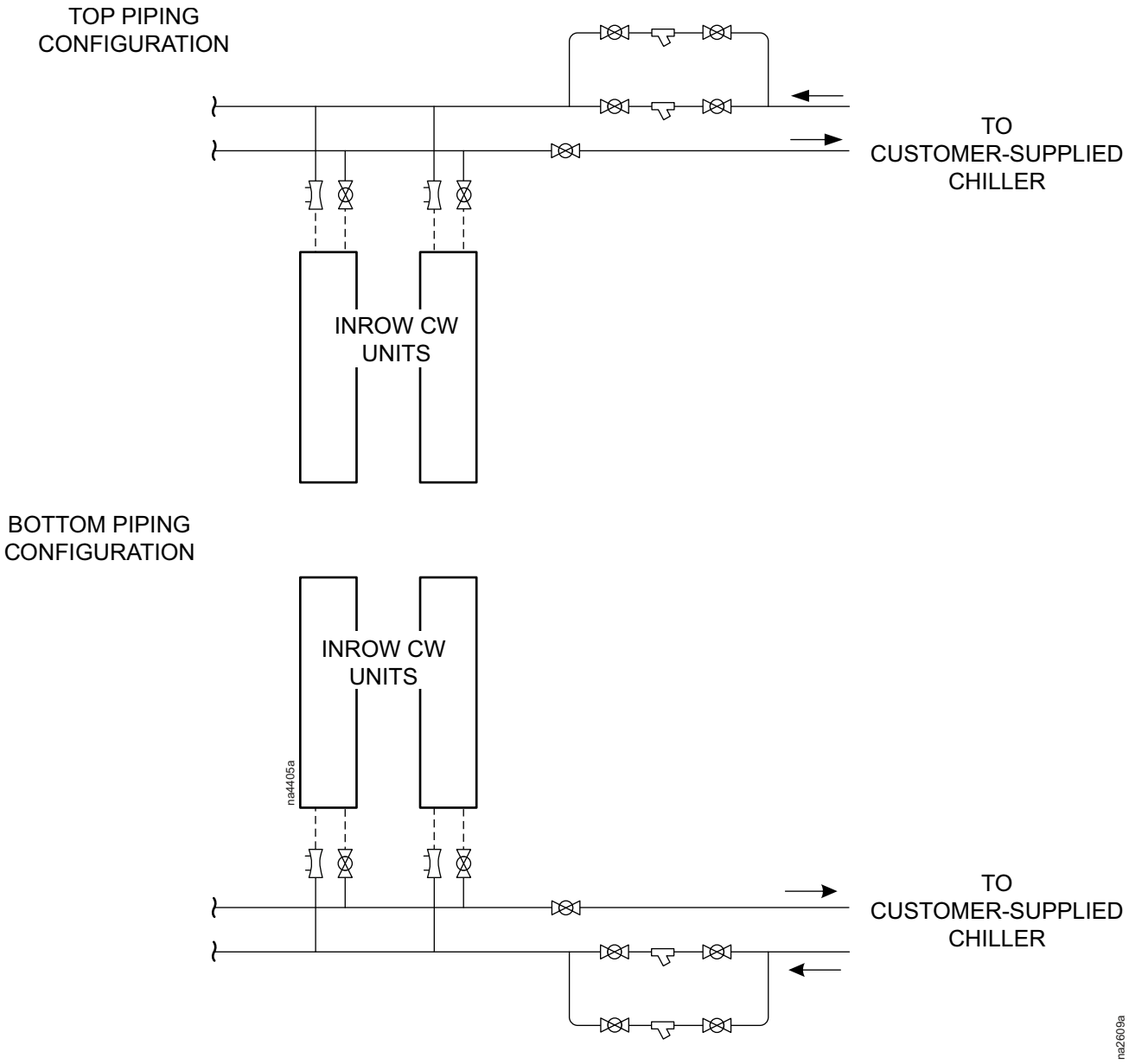
Piping Diagrams

NOTE: Install isolation shutoff valves and particulate strainers in the supply line between the chiller and the CDU. If the system is to be set up as an isolated loop - a chiller and pipe layout, dedicated only to supplying InRow RC units and no other cooling unit - the strainer may be placed in the piping circuit before the pump.

NOTE: Thoroughly flush the system to remove all debris and process chemicals.

NOTE: Top or bottom entry can be chosen individually for each type of connection, i.e., power, condensate drain (ACRC301S), chilled-water supply, and chilled-water return. Top piping configurations have the same valves and strainers as bottom piping configurations.

Connect with Customer Supplied Chiller



Flex hose or copper tubing

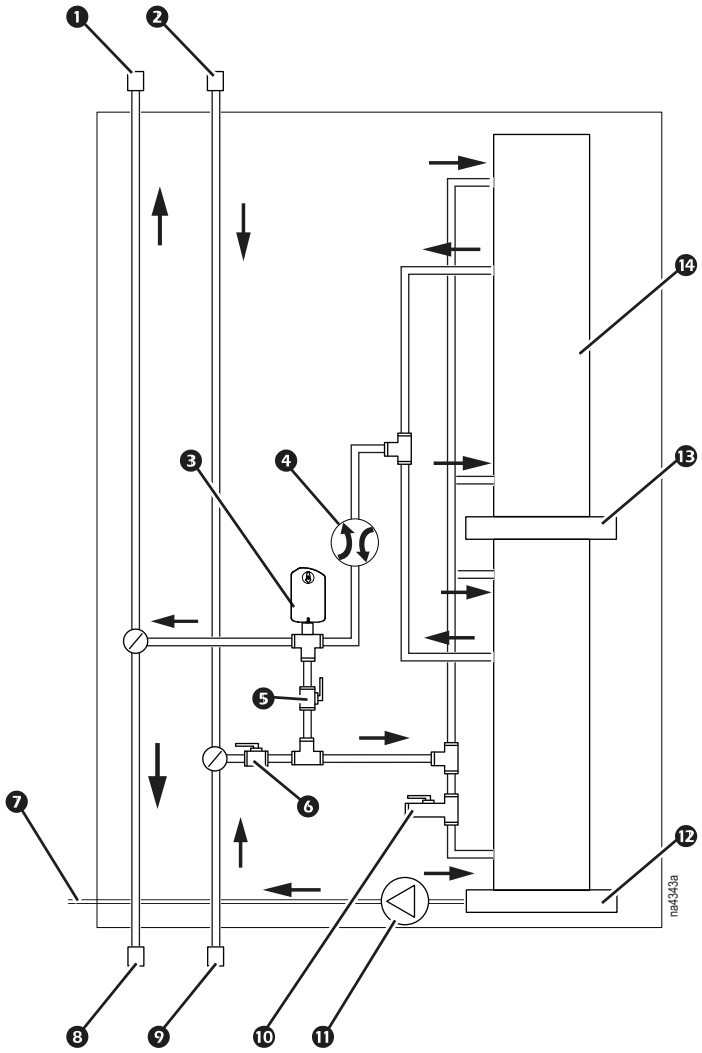
Copper tubing

Circuit Setter

Strainer with 20 mesh screen with opening size of 865 micron, (field installed)

Isolation shutoff valve (field installed)

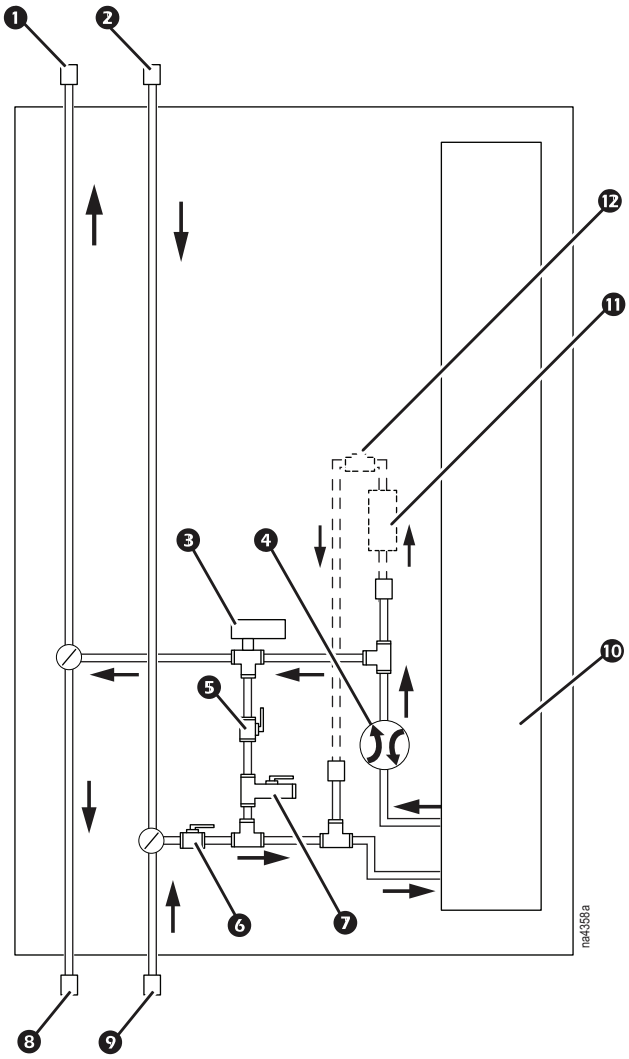
Internal Piping Diagram—ACRC301S



Item	Description	Item	Description
1	Outlet water union (top piping option)	8	Outlet water union (bottom piping option)
2	Inlet water union (top piping option)	9	Inlet water union (bottom piping option)
3	3-way actuator control valve—1 1/4 in.	10	Drain valve
4	Flow meter	11	Condensate pump
5	Bypass shutoff ball valve—3/4 in.	12	Bottom condensate pan

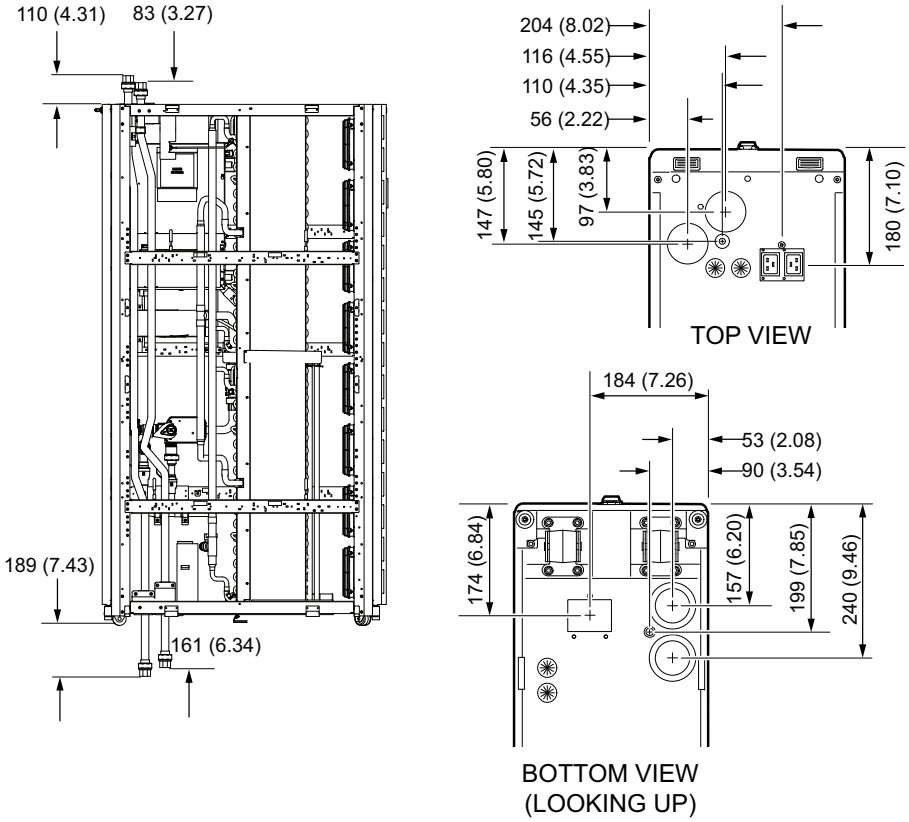
Item	Description	Item	Description
❻	Inlet shutoff valve—1 in.	⓭	Top condensate pan
❼	Condensate drain	⓮	Coil

Internal Piping Diagram—ACRC301H

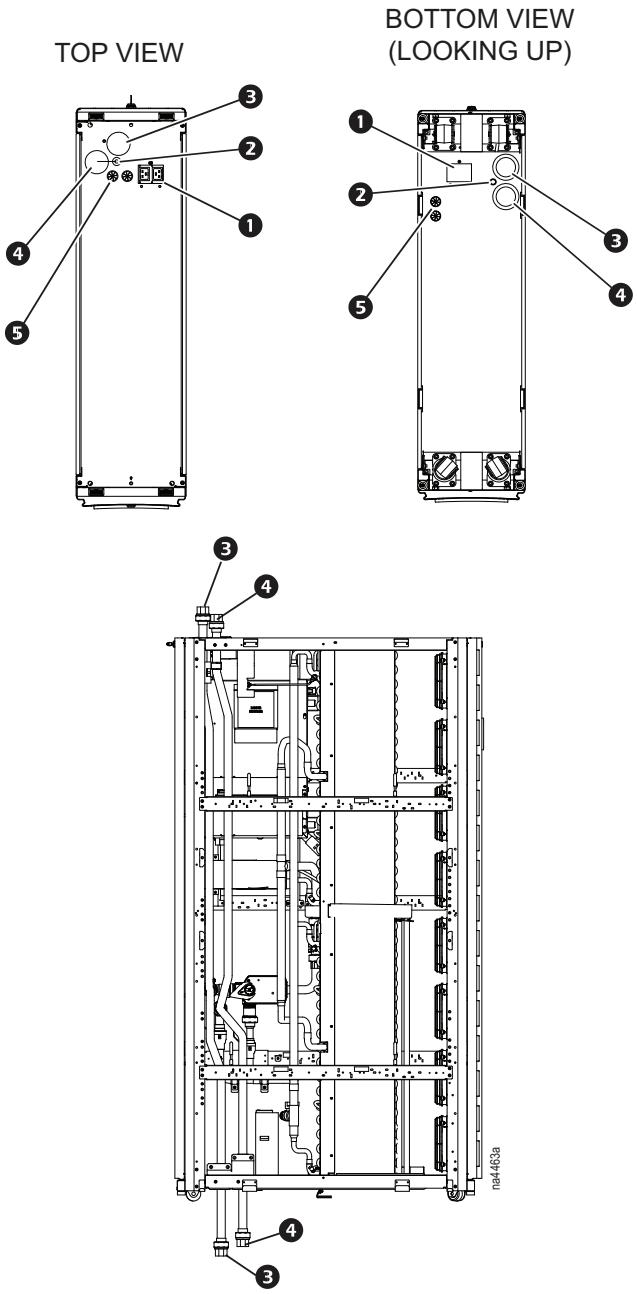


Item	Description	Item	Description
❶	Outlet water union (top piping option)	❼	Drain valve
❷	Inlet water union (top piping option)	❽	Outlet water union (bottom piping option)
❸	3-way actuator control valve—1 1/4 in.	❾	Inlet water union (bottom piping option)
❹	Flow meter	❿	Coil
❺	Bypass shutoff valve—1 in.	⓫	Circulation pump (optional)
❻	Inlet shutoff valve—1 1/4 in.	⓬	Circulation pump check valve (optional)

Piping and electrical access locations—ACRC301S

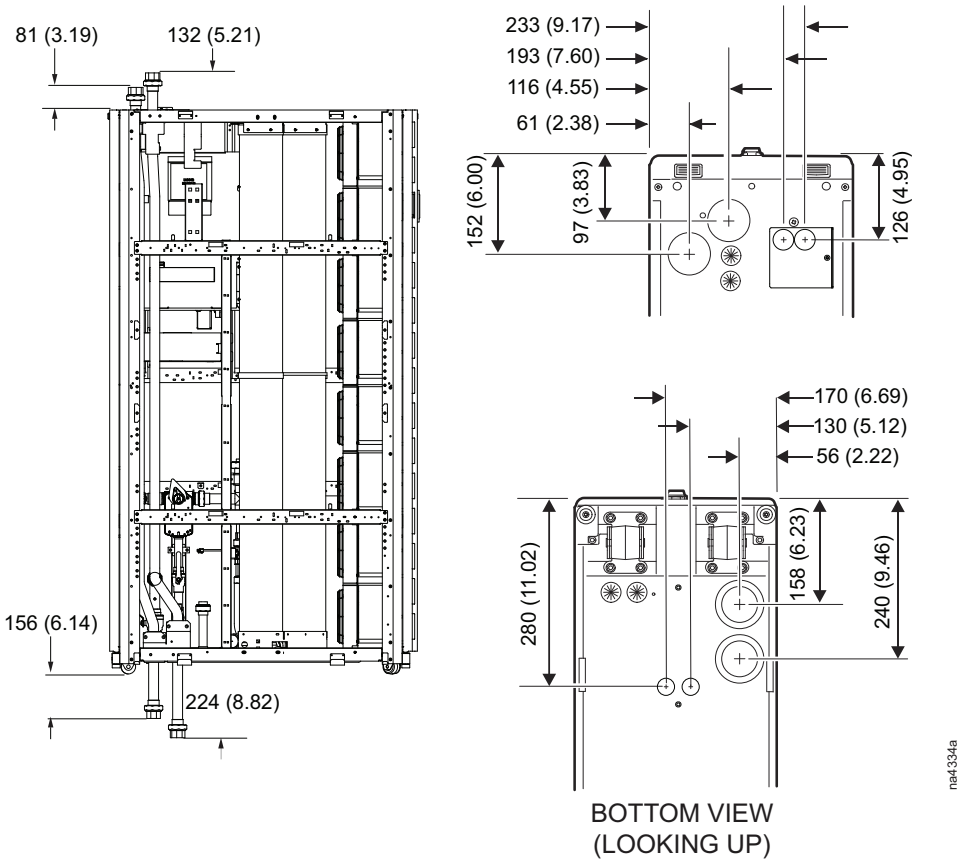


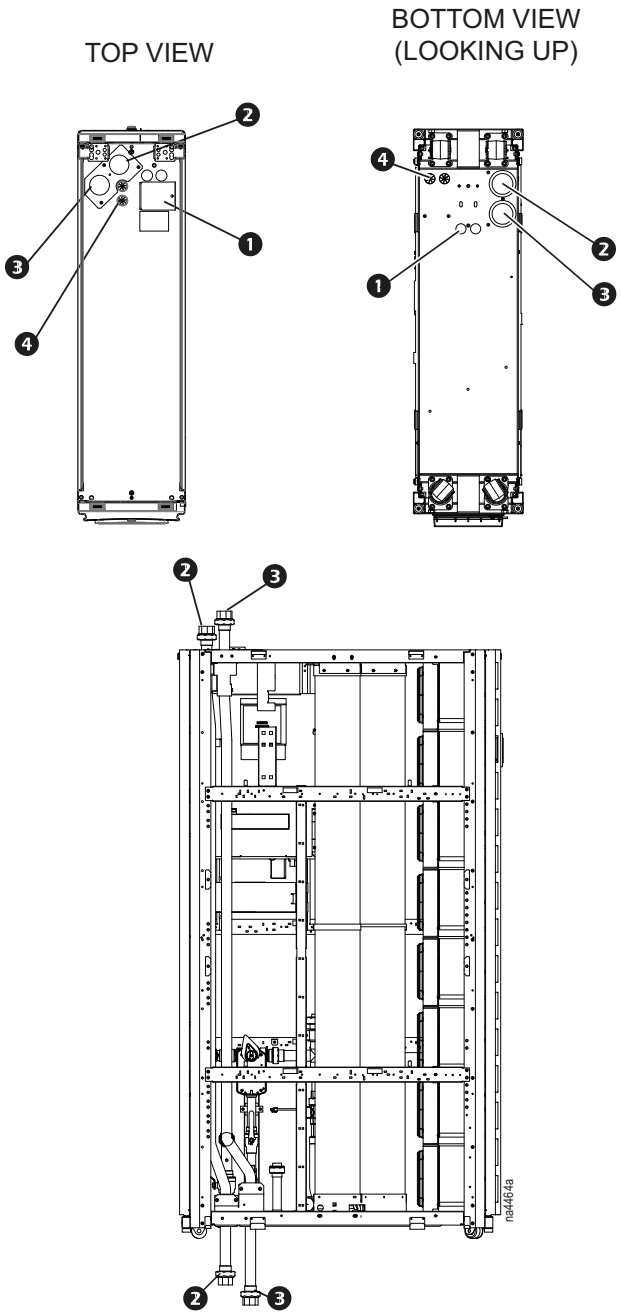
* Dimensions are shown in mm (in.).



Item	Description
❶	Power connections
❷	Condensate line—0.25 in. ID/0.38 in. OD
❸	1-in. NPT female return pipe (outlet)
❹	1-in. NPT female supply pipe (inlet)
❺	Low voltage input wiring.

Piping and electrical access locations—ACRC301H

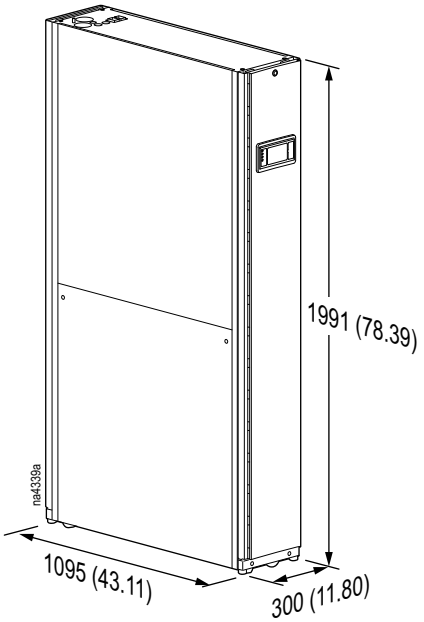




* Dimensions are shown in mm (in.).

Item	Description
❶	Power connections
❷	1 1/4-in. NPT female return pipe (outlet)
❸	1 1/4-in. NPT female supply pipe (inlet)
❹	Low voltage input wiring

Weights and dimensions



* Dimensions are shown in mm (in.).

Net weight (cooling unit only)

- ACRC301S = 184 kg (406 lb)
- ACRC301H = 210 kg (463 lb)

Operating weight (cooling unit only)

- ACRC301S = 192 kg (423 lb)
- ACRC301H = 220 kg (485 lb)

Installation

⚠DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

Turn off all power supplying this equipment before working on the equipment. All electrical work must be performed by qualified personnel. Practice Lockout/Tagout procedures. Do not wear jewelry when working with electrical equipment.

Failure to follow these instructions will result in death or serious injury.

⚠WARNING

MOVING PARTS HAZARD

- Do not open doors and panels if the equipment is operating

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠WARNING

HAZARD TO EQUIPMENT OR PERSONNEL

All work must be performed by Schneider Electric qualified personnel.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠WARNING

TIP HAZARD

- Use two or more people at all times to move or turn this equipment.
- Always push, pull, or turn while facing the front and rear of this equipment - never push, pull, or turn while facing the sides of the equipment.
- Slowly move this equipment across uneven surfaces or door thresholds.
- Lower leveling feet to the floor when this equipment is stationary.
- Lower the leveling feet and attach joining brackets to adjacent racks when the equipment is in its final location.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

⚠CAUTION

UNPROTECTED PARTS

Be careful when placing doors when removed from the equipment. Spring latches are easily damaged.

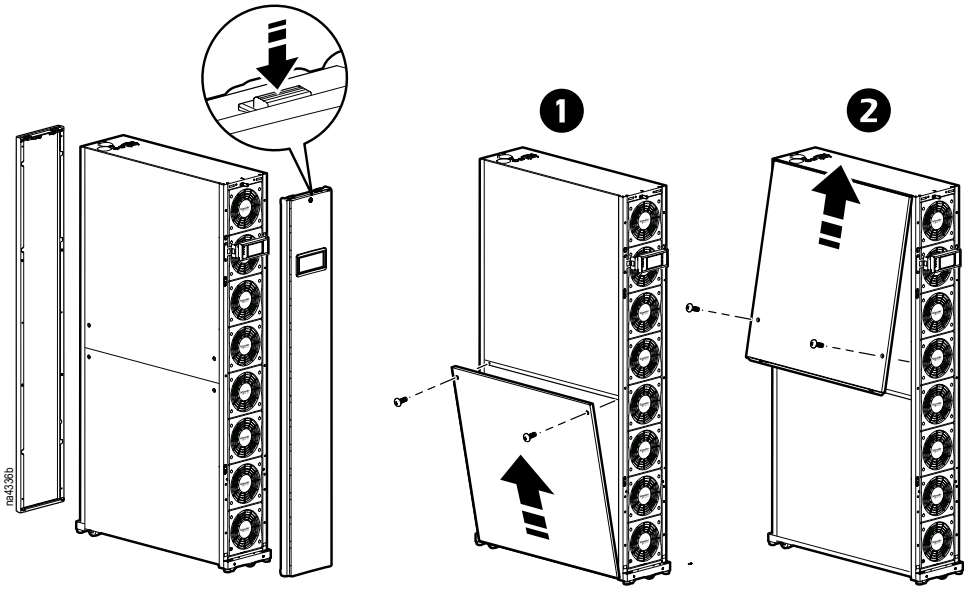
Failure to follow these instructions can result in injury or equipment damage.



Refer to the documentation that is supplied with other equipment for instructions on how to install that equipment.

How to Remove Cover Panels

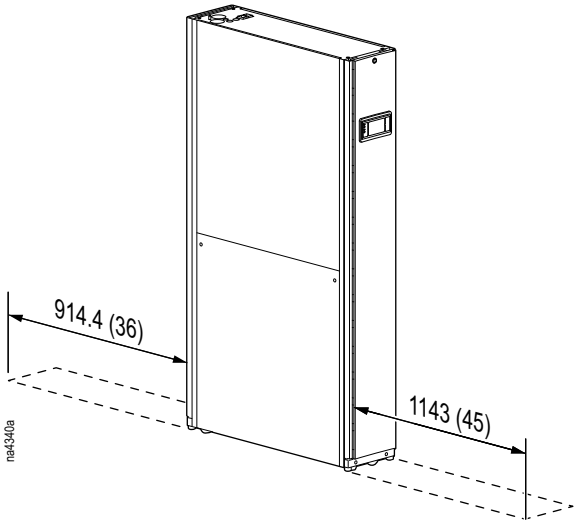
The front and rear panels must first be unlocked before they can be removed. Two keys are provided with the unit.



Positioning the Cooling Unit

Service Access

An area of 1143 mm (45 in.) of clear floor space in front and 914.4 mm (36 in.) in the rear of the cooling units are required for service.



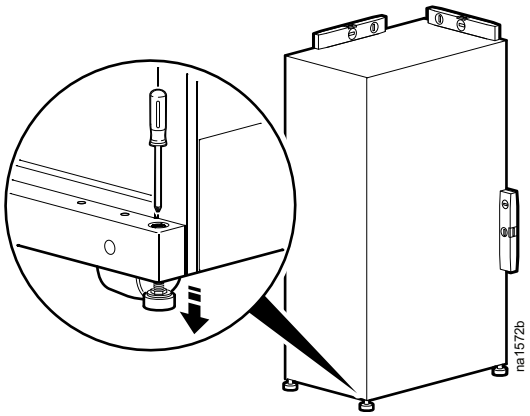
Leveling

The leveling feet provide a stable base if the floor is uneven but cannot compensate for a badly sloped surface.

Once the cooling unit is in its intended location, use a screwdriver to turn each leveling foot until it makes contact with the floor. Adjust each foot until the cooling unit is level and plumb. The casters and leveling feet can be removed to allow the cooling unit to rest directly on the floor.

NOTE: Front and rear panels will need to be removed to access the leveling screw.

NOTE: The image below is an example only. The unit in use may differ.



NOTE: Alternatively, use a 13-mm open-ended wrench to level the equipment without removing the doors.

Stabilizing the Cooling Unit

⚠⚠DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

Turn off all power supplying this equipment before working on the equipment. All electrical work must be performed by qualified personnel. Apply Lockout/Tagout procedures. Do not wear jewelry when working with electrical equipment.

Failure to follow these instructions will result in death or serious injury.

⚠WARNING

HAZARD OF EQUIPMENT FALLING OVER

- Use two or more persons at all times to move or turn this equipment.
- Always push, pull, or turn while facing the front and rear of this equipment. Never push, pull, or turn while facing the sides of this equipment.
- Slowly move this equipment across uneven surfaces or door thresholds.
- Lower leveling feet to floor when this equipment is at rest.
- Lower leveling feet and attach joining brackets to adjacent racks when this equipment is in final position.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Bolt-Down Kit

To prevent the cooling unit from moving (if it is not joined with an enclosure), use the included bolt-down kit (AR7701). Follow the installation instructions included with the kit.

An optional seismic kit (ACAC0MVS1) can be used to secure the cooling unit and prevent it from moving.

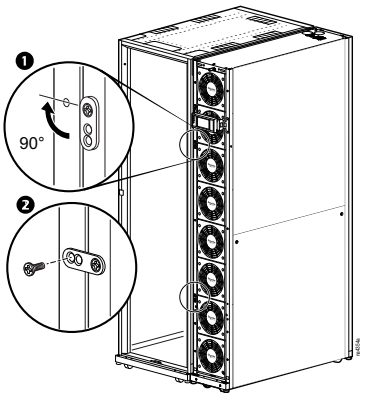
Refer to the installation instruction included in the kit for more detailed information.

Joining to NetShelter™ SX Enclosures

Baying brackets are installed on the unit, two on the front and two on the rear. Each bracket is designed to accommodate both 24 in. or 600 mm enclosure spacing.

NOTE: Image is an example only: the unit may differ.

1. Loosen the attachment screw.
2. Rotate the brackets 90°.
3. Install a provided Phillips M5 screw through the bracket and into the adjoining enclosure.
4. Retighten the attachment screw.



Mechanical Connections

⚠⚠DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

- Apply appropriate personal protective equipment (PPE) and follow safe electrical work practices.
- This equipment must be installed and serviced by qualified and trained personnel only.
- Turn off all power supplying this equipment before working on or inside the equipment.
- Replace all devices, doors, and covers before turning on power to this equipment.

Failure to follow these instructions will result in death or serious injury.

⚠WARNING

HAZARD OF EQUIPMENT FALLING OVER

- Use two or more persons at all times to move or turn this equipment.
- Always push, pull, or turn while facing the front and rear of this equipment. Never push, pull, or turn while facing the sides of this equipment.
- Slowly move this equipment across uneven surfaces or door thresholds.
- Lower leveling feet to floor when this equipment is at rest.
- Lower leveling feet and attach joining brackets to adjacent racks when this equipment is in final position.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

Piping

The coolant water quality must comply with local standards or at least with the guidelines below.

Cooling Water Quality Guidelines for APC Cooling Units

1 ppm = 1 mg/kg = 1 mg/l, CFU - colony forming unit

Water Characteristics	Range	Water Characteristics	Range
Alkalinity (HCO_3^-) Hydrogen Carbonate	70 – 300 ppm	Hardness as $CaCO_3$	30 – 200 ppm
Alkalinity / Sulfate: HCO_3^- /SO_4^{2-}	>1.0	Hydrogen Sulfide (H_2S)	< 0.05 ppm
Aluminum (Al)	< 0.2 ppm	Iron (Fe)	< 0.2 ppm
Ammonia (NH_3)	< 2 ppm	Magnesium (Mg)	< 50 ppm
Ammonium (NH_4)	< 2 ppm	Manganese (Mn)	< 0.1 ppm
Bacteria: aerobic	< 1000 CFU/ml	Molybdenates (MoO_4^{2-})	< 30 ppm
anaerobic	< 1000 CFU/ml	Nitrate (NO_3)	< 100 ppm
heterotrophic colony count	< 100,000 CFU/ml	Oil and grease	< 1 ppm
legionella species	< 10 CFU/ml	Oxygen	< 0.02 ppm

Cooling Water Quality Guidelines for APC Cooling Units

1 ppm = 1 mg/kg = 1 mg/l, CFU - colony forming unit

Water Characteristics	Range	Water Characteristics	Range
sulfate reducing	< 10 CFU/ml	Particles: size	< 25 microns
pseudomonas	< 10 CFU/100 ml	weight	< 10 ppm
Calcium (Ca)	< 50 ppm	pH: Acid-Alkaline balance	7 – 9
Carbon Dioxide (CO_2): free (aggressive)	< 5 ppm	Phosphate (PO_4^{3-})	< 10 ppm
Carbon acid, free	< 20 ppm	Potassium (K)	< 10 ppm
Chloride (Cl^-)	< 300 ppm	Sodium (Na)	< 10 ppm
Chlorine, free (CL_2)	< 1 ppm	Softness: [Ca^{2+}, Mg^{2+}]/[HCO_3^-]	> 0.5
Copper (Cu)	< 1.0 ppm	Silica (SiO_2) Silicon Dioxide	< 50 ppm
Electrical conductivity	50 – 500 µS/cm	Sulfate (SO_4^{2-})	< 50 ppm
		Sulfide (S^{2-})	0 ppm

Water

Install shutoff valves for routine service and emergency isolation of the cooling unit. When a CDU is not used, it is recommended that you install circuit setters to regulate the chilled water flow for each InRow RC air conditioner.

Refer to Piping Diagrams, page 26.

Layout and Piping Considerations

Fluid pipes are not allowed directly above electrical cooling units and must be kept separate from any electrical runs or wiring. All fluid piping must be installed above the aisles. If any piping makes a turn, or must be routed over electrical equipment, there must be a drip tray under the pipe that will protect the equipment from condensation and leaks.

Insulation

Insulate water lines to protect personnel, protect components from condensate, and to minimize condensation

See Insulation, page 44.

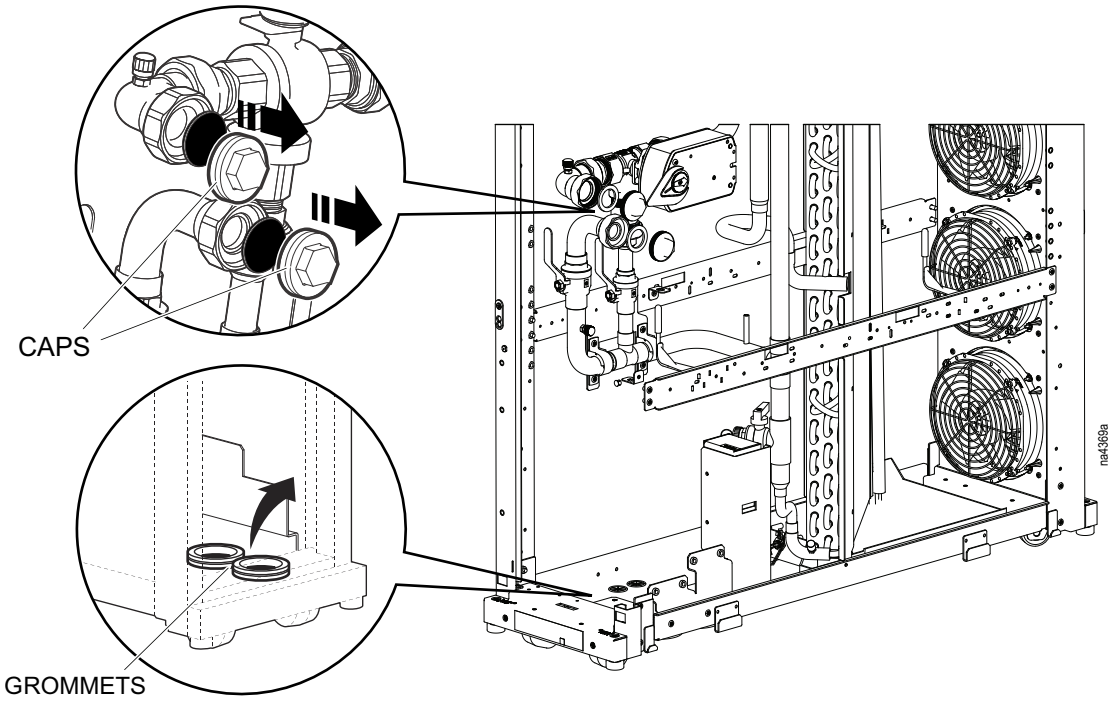
Supply and Return Piping

NOTE: This section covers a bottom piping configuration only. The ACRC301S model is shown in the illustrations; the process will be the same for ACRC301H units.

NOTE: Before installing pipes, use the service valves to discharge gas from the unit. For service valve locations, see Internal Components—ACRC301S, page 19 or Internal Components—ACRC301H, page 20.

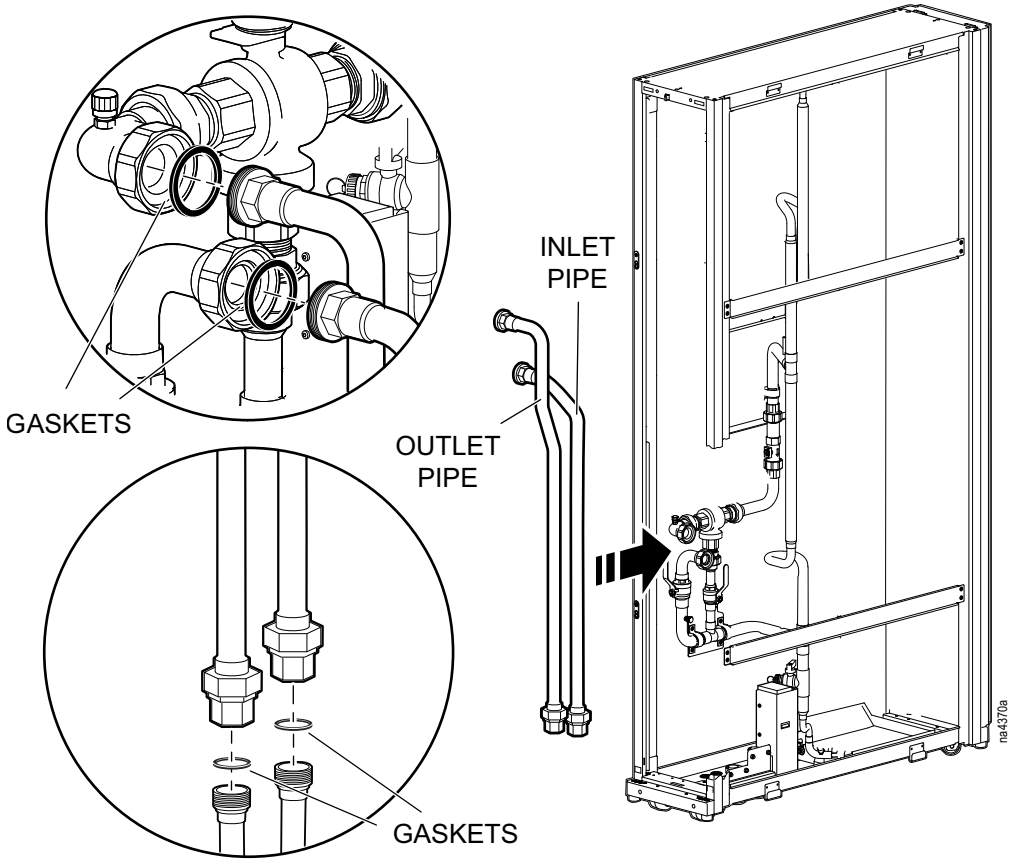
For upper piping configurations, see the instructions that come with the upper piping kit.

1. Remove the pipe connection caps and grommets from the bottom pipe connections.

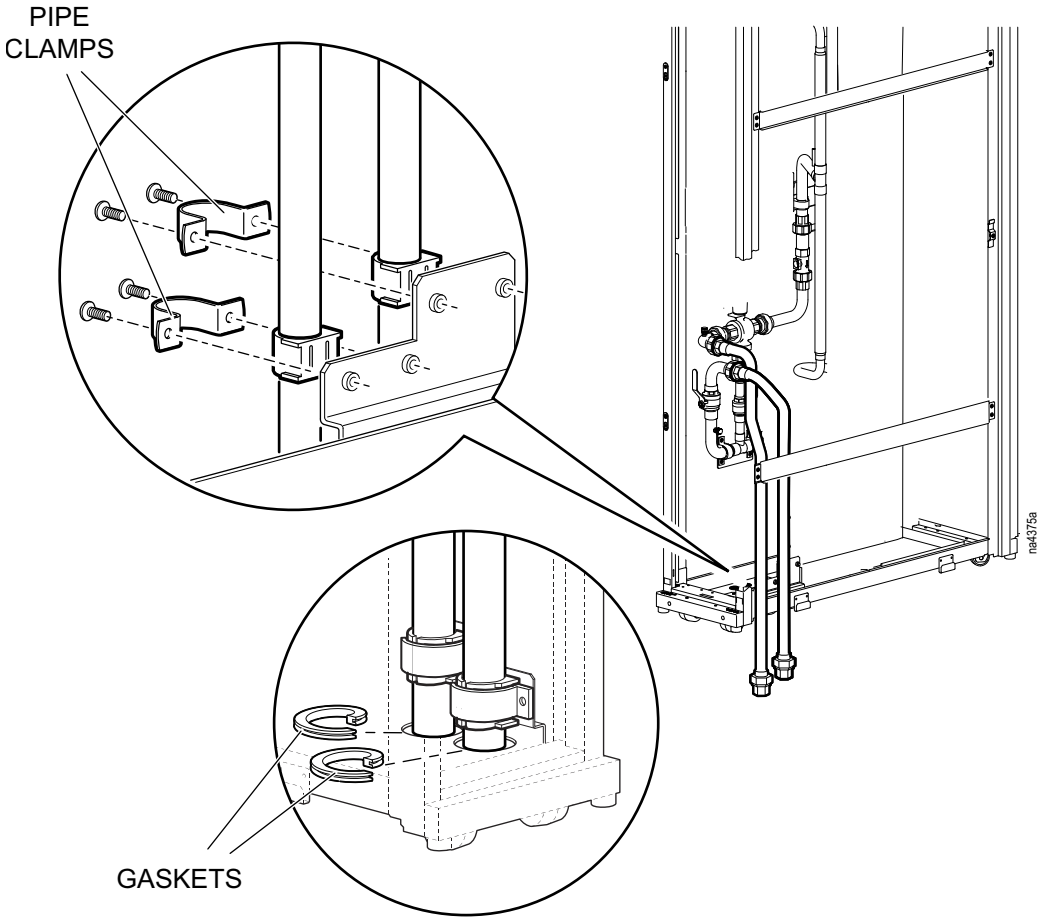


2. Place the lower piping insulation on the pipes.

3. Install the provided pipe gaskets and piping. Tighten connections to 40 N*m (29 ft/lb).



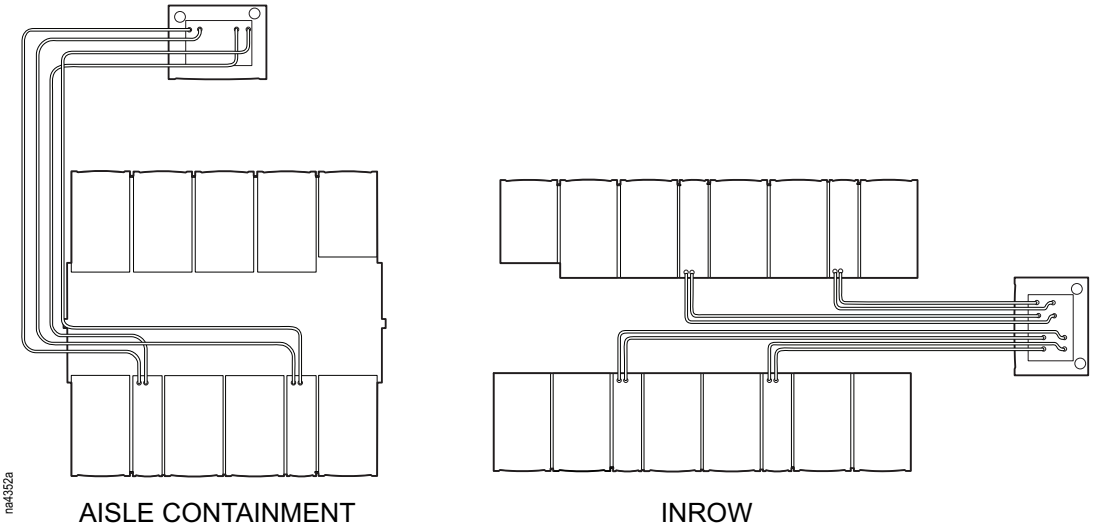
4. Install lower pipe clamps and re-install the grommets around the pipes.



5. Install the insulation around the pipe clamps.

Main Water Connections

System Piping Diagram

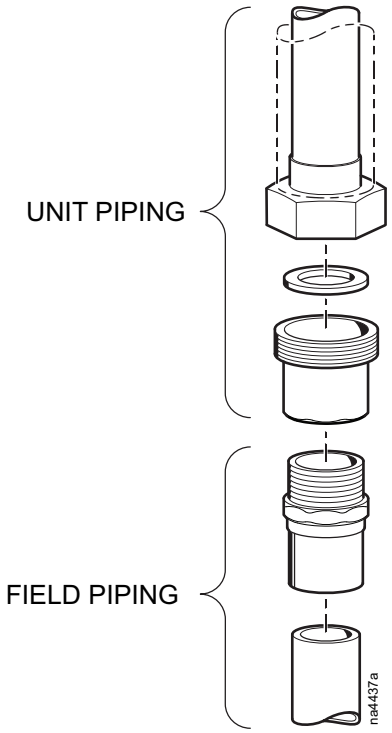


1. Route all piping to the cooling unit. Be sure to follow all local and national codes.

   NOTE: Circuit setters are recommended to regulate the flow of chilled water to each cooling unit, except for the units installed with EPIV.

2. Make the connection on the entering and exiting water lines:
   - Use thread sealant and thread sealing tape in accordance with local and national codes.
   - Use two properly sized open end wrenches to tighten the connections.

BOTTOM PIPING SHOWN

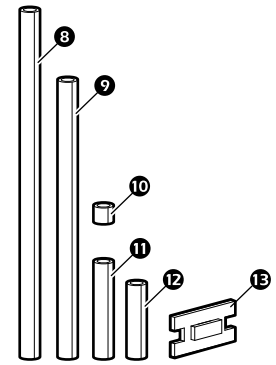


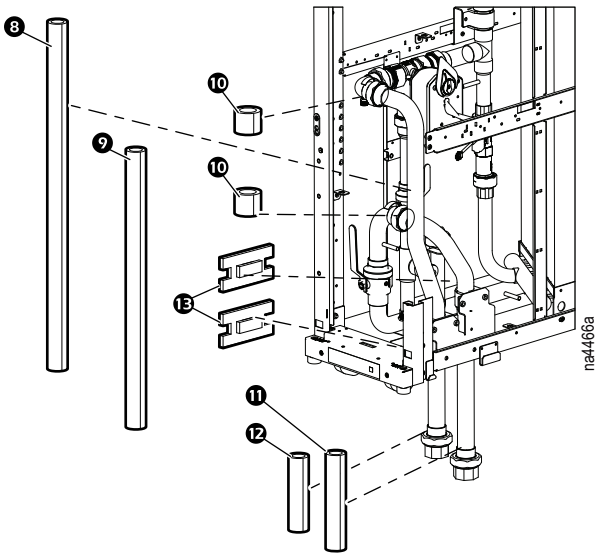
NOTE: The figure shows field piping installation with a field-supplied PEX fitting. Rigid pipe fittings are handled similarly.

Insulation

Use only approved insulation (closed cell elastomeric insulation with sealing seams). Insulation should be 9.52 mm (3/8 in.) wall thickness. All horizontal insulation sections must be installed with seams facing up. Each section of insulation must be glued to the adjacent section. Any insulation sections that must be fitted around piping support clamps (other than supplied clamps) must be glued together to prevent condensation.

See Package Contents—ACRC301S, page 15 and Package Contents—ACRC301H, page 16 for locations of insulation pieces.





NOTE: ACRC301H is shown.

Condensate Pump and Drain Line—ACRC301S

⚠CAUTION

CONDENSATE DAMAGE

- Do not exceed the lift or the run length of the drain system.
- To prevent equipment damage from condensate, do not leave the condensate drain line coiled inside the cooling unit. Route the condensate drain line out the top of the cooling unit before operation.

Failure to follow these instructions can result in injury or equipment damage.

NOTICE

COMPLIANCE REQUIREMENT

The installation must comply with local plumbing codes.

Failure to follow these instructions can result in equipment damage.

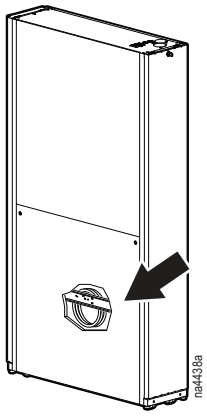
The condensate pump is factory-wired, piped internally to the lower condensate pan, and is capable of moving liquid a maximum of 15.2 m (50 ft), including a maximum lift of 4.9 m (16 ft).

NOTE: Sufficient drain line is supplied to route the drain to the outside of the cooling unit. To route the drain line to a remote drain, an additional drain line may need to be provided.

An on-board condensate high-level float switch is wired into the InRow CW controller for local and remote alarm capabilities.

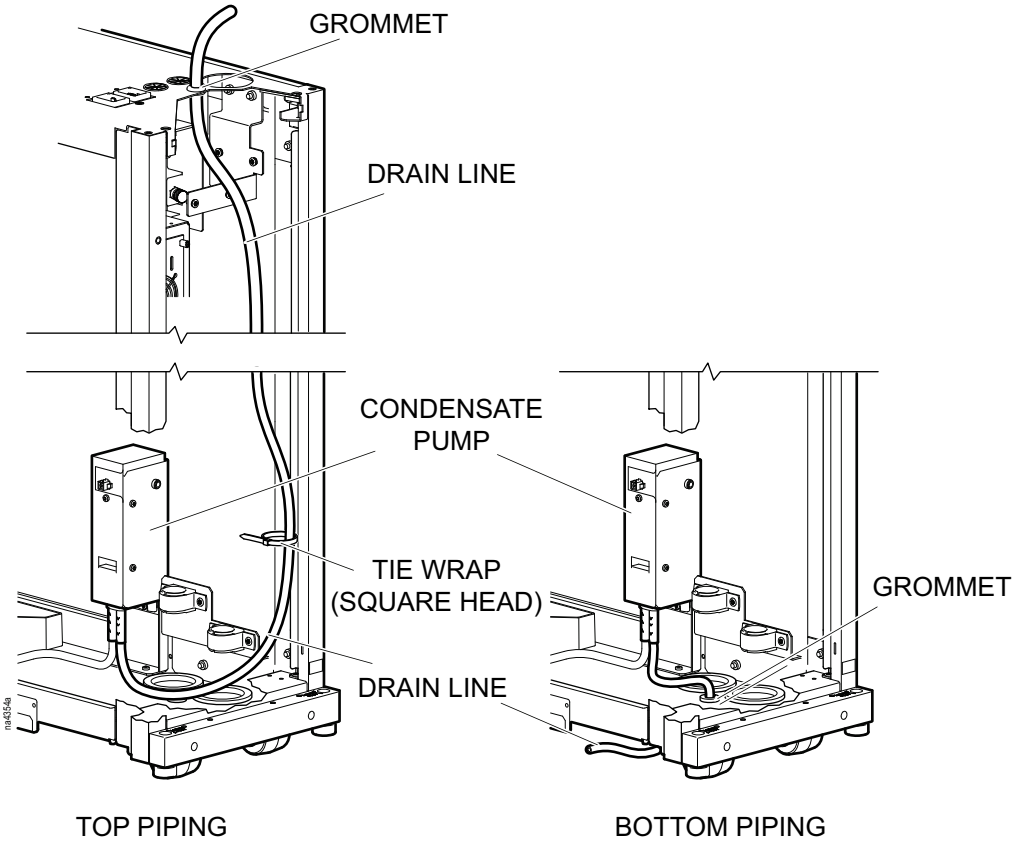
To route the pump drain line:

1. Remove the drain line from the side of the unit.



2. Route the drain line through the top or bottom of the unit. Use the provided rubber grommets.

3. Secure the drain line with tie wraps.



Additional Equipment

Chiller

There are three types of chillers to which the cooling unit can be connected:

- APC size-matched chiller/thermal storage system
- Building chilled-water system
- Existing dedicated chiller

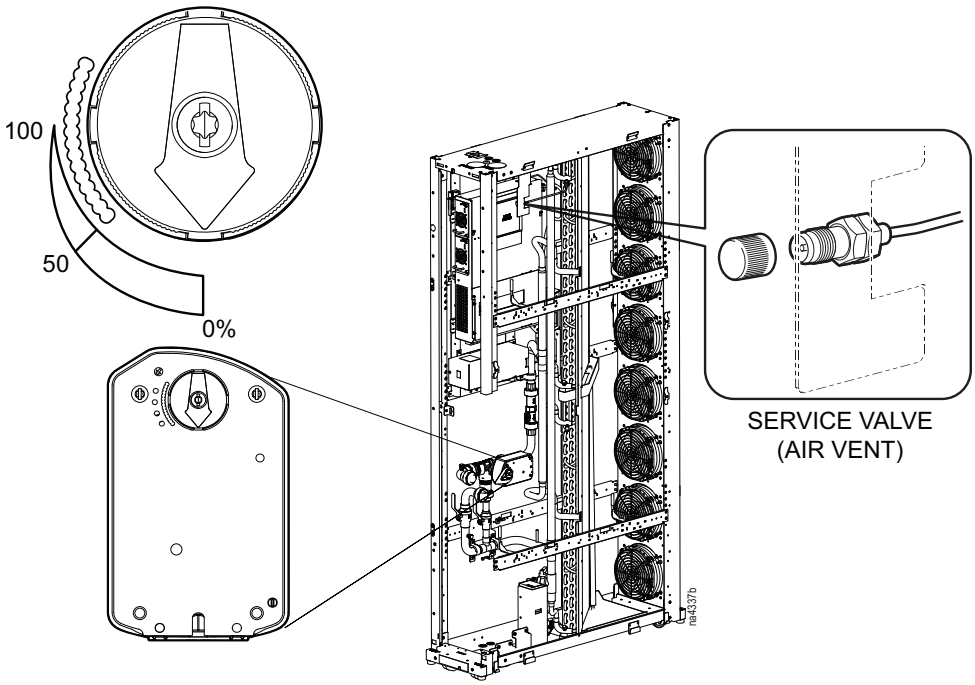
See the chiller documentation for proper installation procedures.

Filling and Purging

When the installation of the piping is complete, begin the filling process.

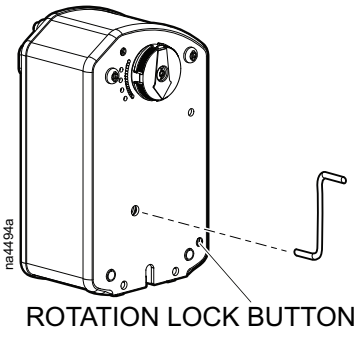
1. Open all isolation valves and allow water to flow through the cooling unit.
2. Manually rotate the chilled-water control valve stem to the fully open position.

   NOTE: To manually reposition a spring-return actuator from a locked position, turn the crank 1/8 of a turn to the right, then remove the crank to release



   NOTE: ACRC301S is shown.

   a. Insert the attached crank into the hole located on the front of the actuator.



   b. Rotate the crank in the direction indicated on the label until the valve is in the required position.

   c. Hold the crank in place and, using a Phillips screwdriver, turn the rotation lock button in the direction indicated on the label.

   d. Hold the screwdriver in place, remove the crank, then remove the screwdriver.

3. Open the cap of the service valve and slightly press the needle.
4. Close the cap of the service valve when water begins flowing out of the vent.

At the water supply:

1. Open all valves (no greater than 108l/m [28.5 gpm] for ACRC301S units and 144 l/m [38 gpm] for ACRC301H units), allowing the water supply to reach the highest possible flow to the cooling unit for 45 seconds.
2. Close the valves to a 3.8–11.4 l/m (1–3 gpm) flow for 60 seconds.
3. Open the valves to maximum flow for another 45 seconds.

4. Balance the system to provide the designed flow rate to all cooling units.

Electrical Connections

⚠⚠DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

- Apply appropriate personal protective equipment (PPE) and follow safe electrical work practices.
- This equipment must be installed and serviced by qualified and trained personnel only.
- Turn off all power supplying this equipment before working on or inside the equipment.
- Replace all devices, doors, and covers before turning on power to this equipment.

Failure to follow these instructions will result in death or serious injury.

⚠⚠DANGER

HAZARD OF ELECTRIC SHOCK, EXPLOSION, OR ARC FLASH

Potentially dangerous and lethal voltages exist within this unit. More than one disconnect switch may be required to energize or de-energize this equipment. Observe all cautions and warnings.

Failure to follow these instructions will result in death or serious injury.

⚠⚠WARNING

ELECTRICAL HAZARD

- Electrical service must conform to local and national electrical codes and regulations.
- The equipment must be grounded.

Failure to follow these instructions can result in death, serious injury, or equipment damage.

The following electrical connections are required in the field:

- Primary and secondary feeds
- Communication (A-Link, Building Management System, Modbus)
- Sensors



See the electrical diagram (located on the electrical box) for all electrical connections.



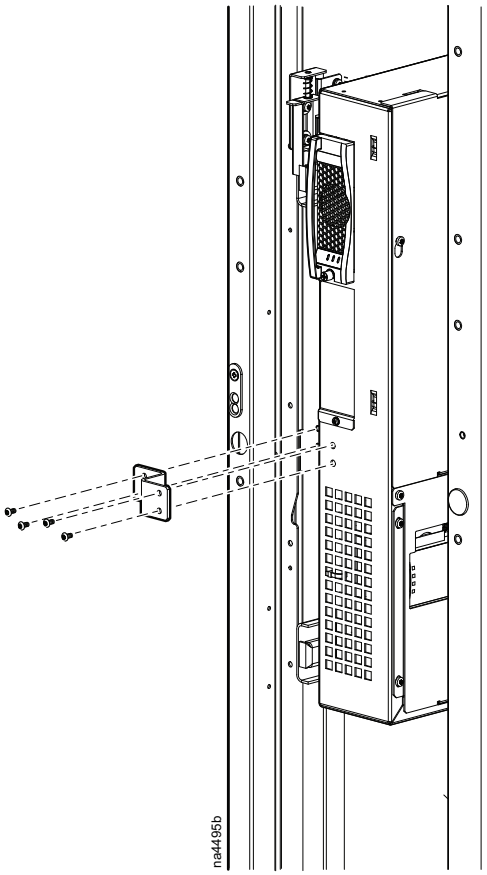
See the equipment name plate for voltage and current requirements.

All low-voltage connections, including data and control connections, must be made with properly insulated wires. Low-voltage wiring must be insulated based on the wiring with which it is routed. The low-voltage connections must have 300-V minimum insulation.

NOTE: A power disconnect switch is required to isolate each unit for maintenance and servicing.

NOTE: Single-phase service is required. Electrical service must conform to national and local electrical codes. The InRow CW unit is grounded through the power cord.

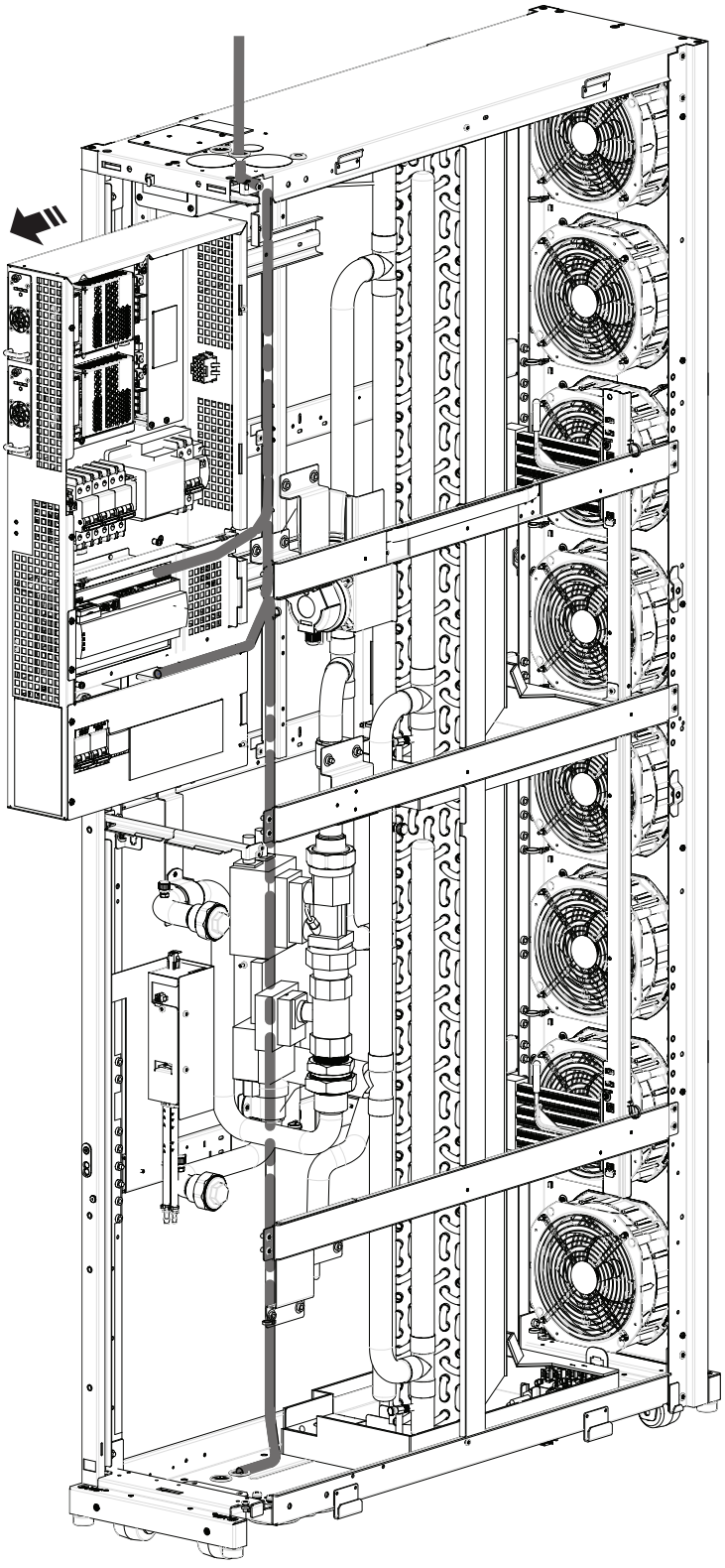
Before making any electrical connections, the shipping bracket must be removed from the electrical box.


na4495b

Communication Connections

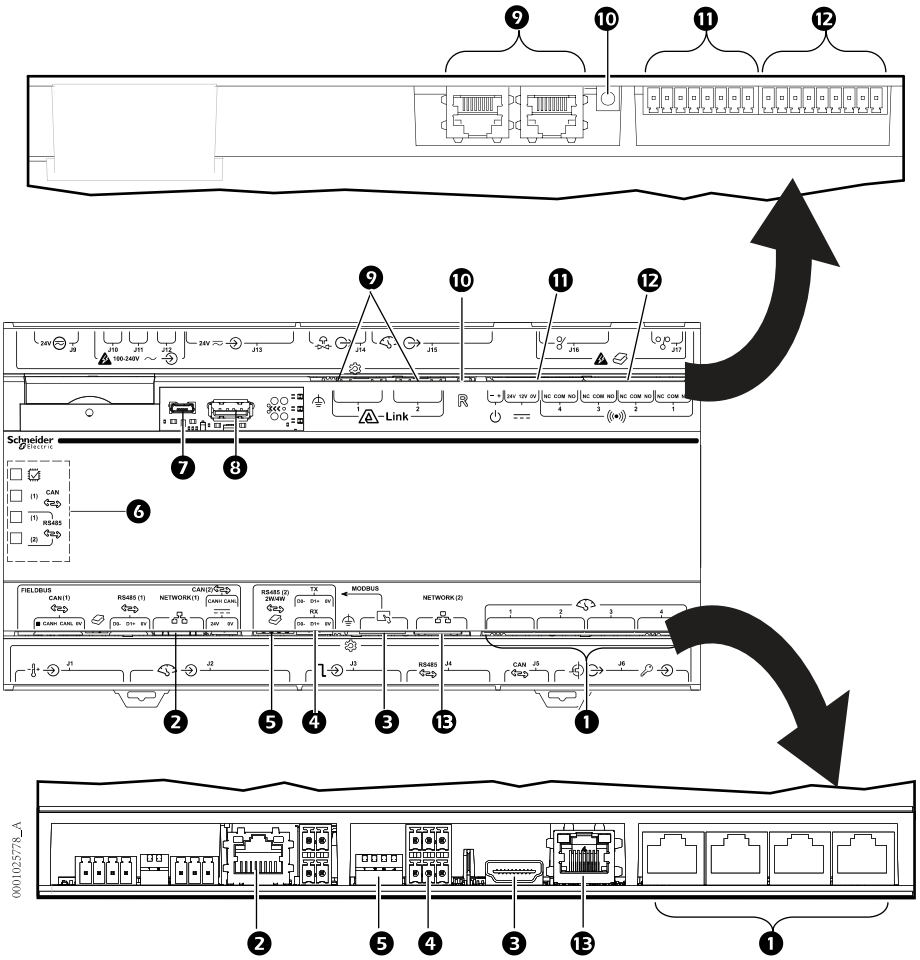
Controller Interface Connections

The wiring harness is routed through the low voltage input wiring at the top or bottom of the unit. For detailed low voltage wiring location, see .



NOTE: When pulling the electric control box out of the unit, pay attention to the cables connected at the back. Leave some slack.

NOTE: Some holes are pre-drilled on the unit slide brackets or support side rail brackets. Select appropriate holes to secure the wires according to the actual situation.



Item	Description	Item	Description
❶	Universal sensor ports	❽	USB host port
❷	Network connection	❾	A-Link port
❸	Touchscreen display connection	❿	Reset button
❹	Modbus connection	⓫	Output relay 4/standby input
❺	Modbus configuration switches	⓬	Output relay 1–3
❻	Processor status LED	⓭	Not in use
❼	USB device port		

A-Link Connections

NOTE: All input and output connections should be wired as Class 2 circuits.

The A-Link bus connection allows a maximum group of 20 units to communicate with each other.

To enable the units to function as a group, link them together using a standard pin-out CAT-5 cable with RJ-45 connectors. The A-Link bus must be terminated at the first and last unit installed in the group. An A-Link terminator is supplied with each unit.

NOTICE
CAT 5 PINOUT Devices connected on the A-Link ports should use a standard pin-out (1-1, 2-2, 3-3, 4-4, 5-5, 6-6, 7-7, 8-8) CAT5 cable only. **Failure to follow these instructions can result in equipment damage.**

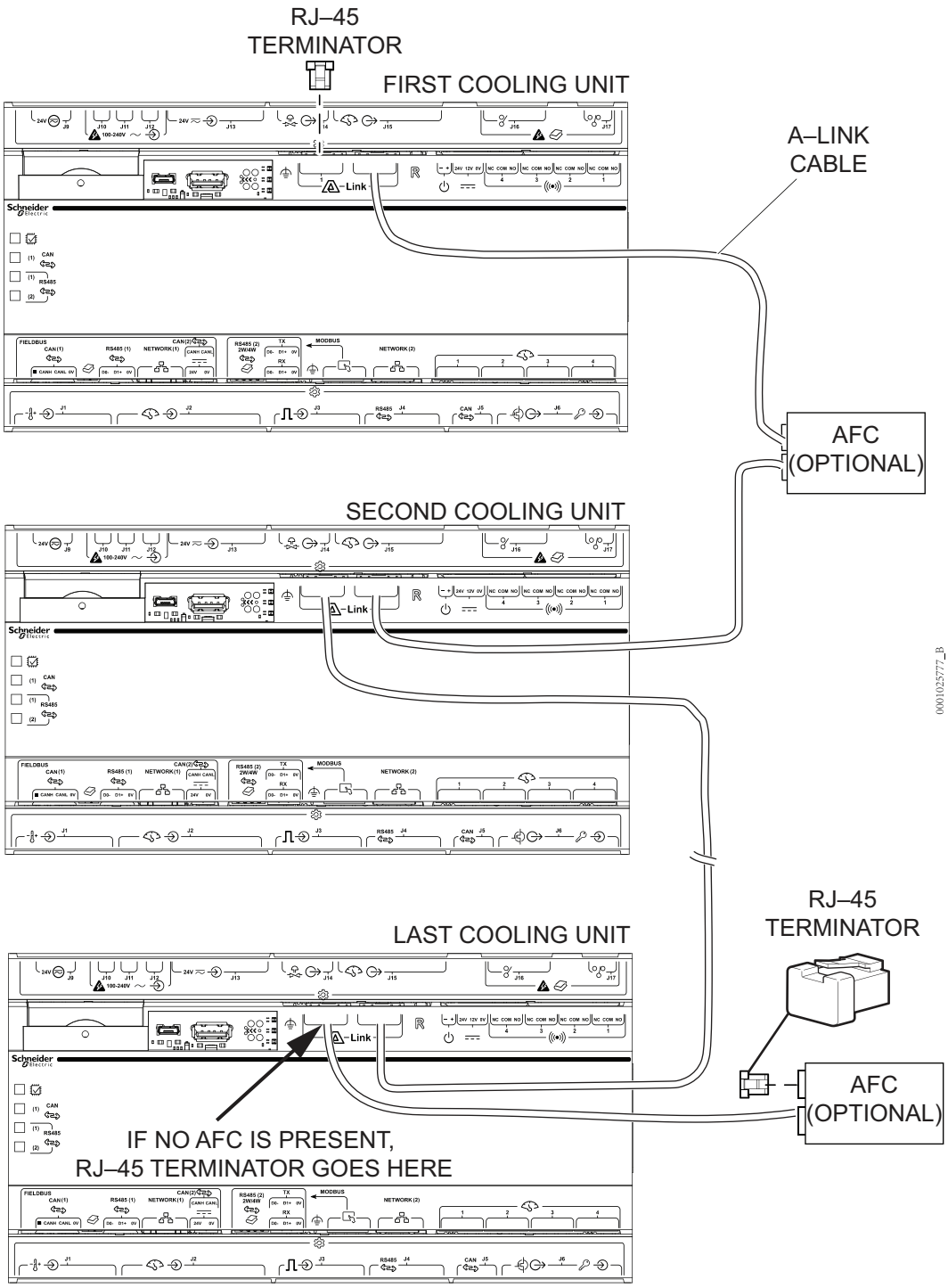
NOTE: The maximum wire length for the entire group must not exceed 500 m (1640 ft).

Active Flow Controllers (AFC)

If AFC units are installed in the cooling group in a HACS or CACS environment, they are connected to the A-Link bus as shown.

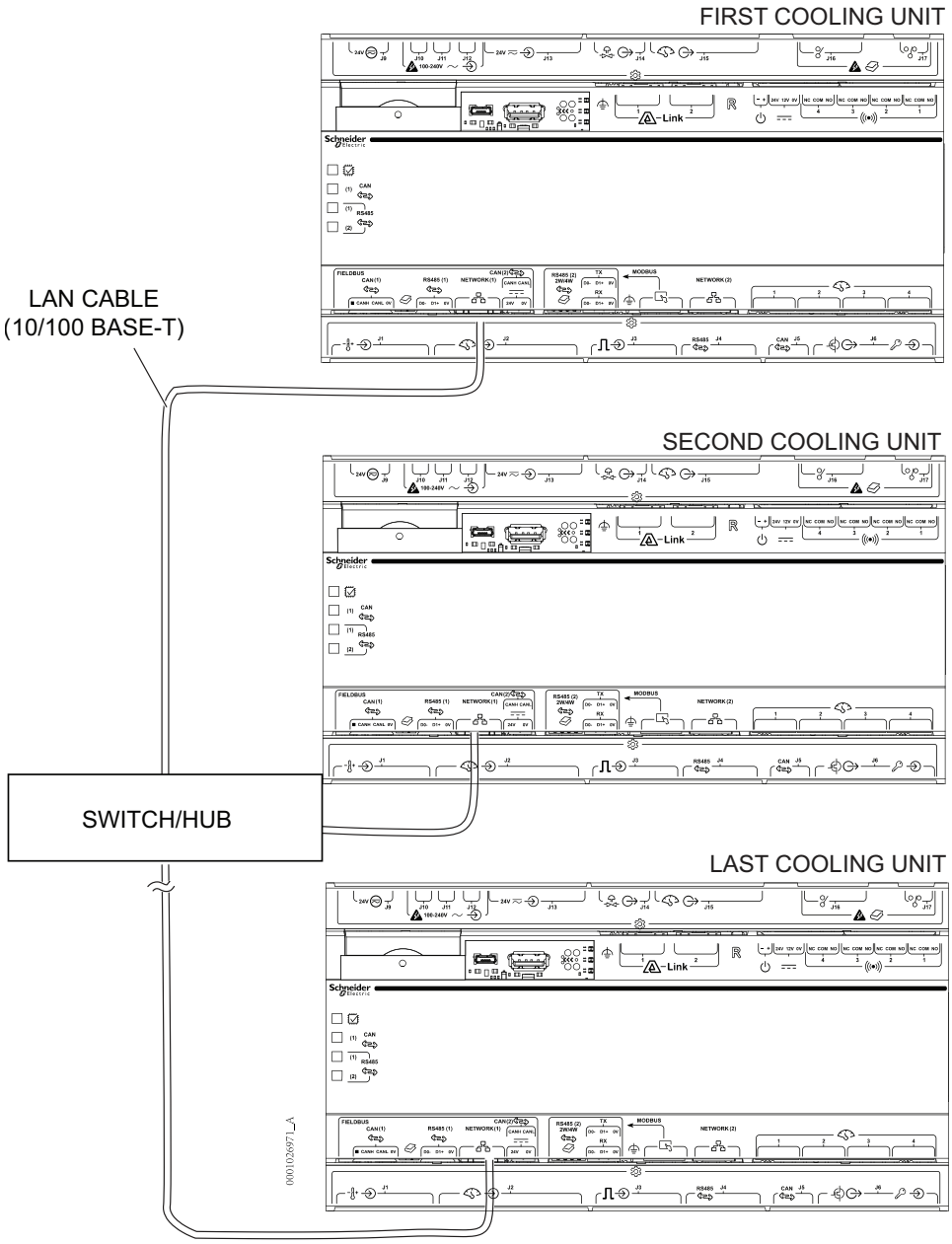


See the *AFC Installation Manual* for more information.


RJ–45
TERMINATOR
FIRST COOLING UNIT
A–LINK
CABLE
AFC
(OPTIONAL)
SECOND COOLING UNIT
0001025777_B
RJ–45
TERMINATOR
LAST COOLING UNIT
AFC
(OPTIONAL)
IF NO AFC IS PRESENT,
RJ–45 TERMINATOR GOES HERE

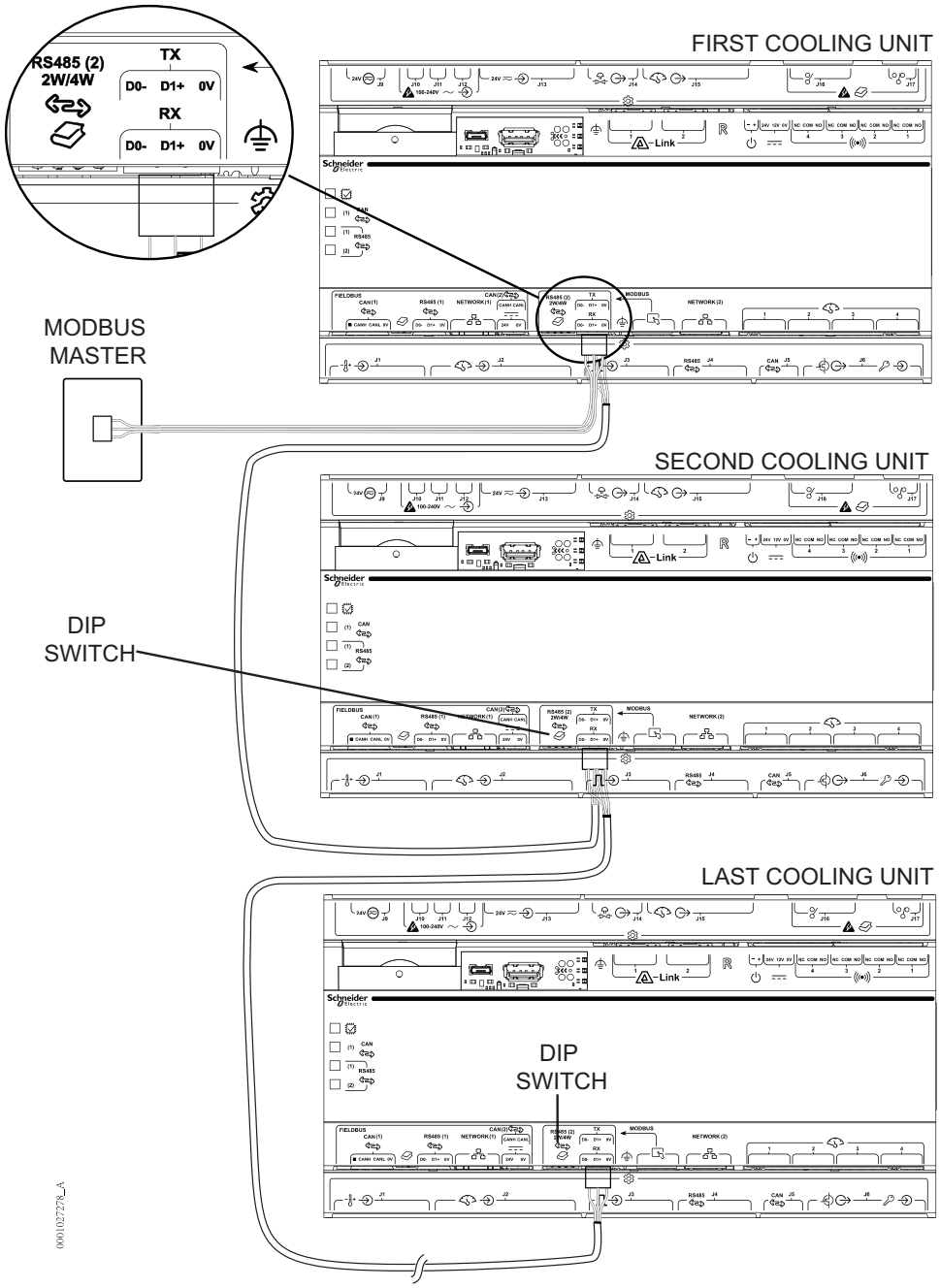
Network Connection

The network connection enables the communication between the BMS and the controllers of the units, utilizing industry-standard protocols such as SNMP, Modbus TCP , and BACnet for seamless integration and real-time monitoring.

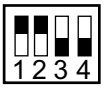
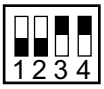
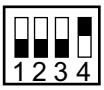


Modbus

The image below shows the BMS communicates with the controllers of the units through the Modbus RTU protocol via serial connections RS-485.

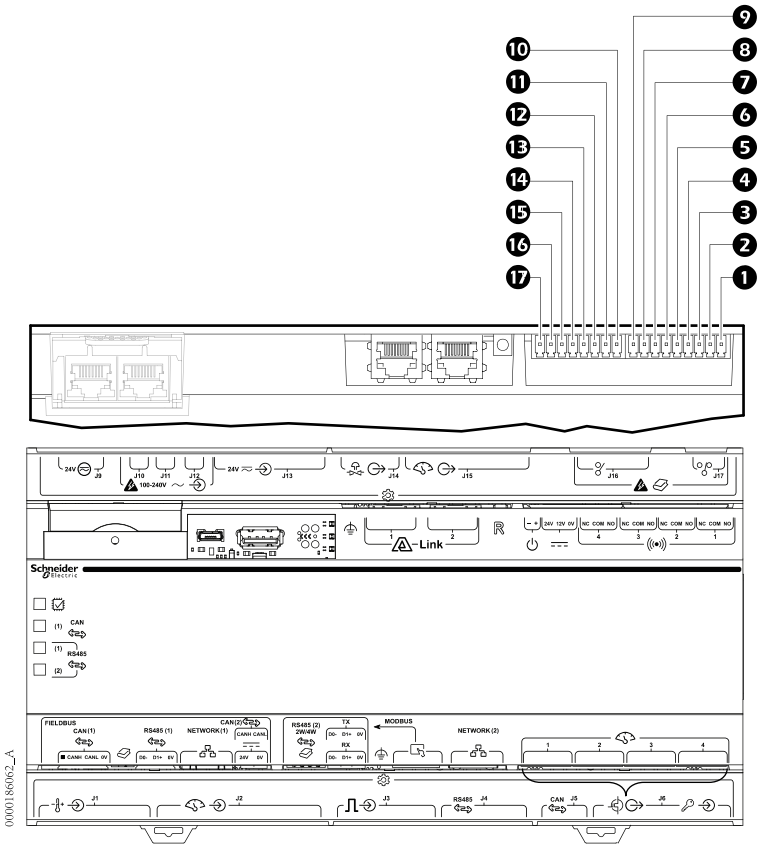

RS485 (2)
2W/4W
TX
D0- D1+ 0V
RX
D0- D1+ 0V
FIRST COOLING UNIT
MODBUS
MASTER
SECOND COOLING UNIT
DIP
SWITCH
LAST COOLING UNIT
DIP
SWITCH
0001027278_A

Modbus Switch Configuration

Position	Status	DIP Switches	Mode
1 and 2	Open (up)		4-wire Modbus (needs TXD0-, TXD1+, RXD0-, RXD0+, and COM connected to Modbus master)
1 and 2	Closed (down)		2-wire Modbus (needs TXD0-, TXD1+, and COM, or RXD0, RXD0+, and COM)
3 and 4	Open (up)		2-wire or 4-wire Modbus (no termination)
3 and 4	Closed (down)		4-wire Modbus (termination enabled)
3	Closed (down)		2-wire Modbus (termination enabled, position 4 must remain in the Open (up) position in 2-wire mode, or double termination will occur)

Output Relays and Standby Input

The wiring harness for output relays and standby input are routed through the low voltage input wiring at the top or bottom of the unit. For detailed low voltage wiring location, see .



Item	Name	Description
❶	Output relay 1 N.O.	Normally Open contact
❷	Output relay 1 COM	Common contact
❸	Output relay 1 N.C.	Normally Closed contact
❹	Output relay 2 N.O.	Normally Open contact
❺	Output relay 2 COM	Common contact
❻	Output relay 2 N.C.	Normally Closed contact
❼	Output relay 3 N.O.	Normally Open contact
❽	Output relay 3 COM	Common contact
❾	Output relay 3 N.C.	Normally Closed contact
❿	Output relay 4 N.O.	Normally Open contact
⓫	Output relay 4 COM	Common contact
⓬	Output relay 4 N.C.	Normally Closed contact
⓭	Ground	Ground reference, typically connected to the negative connection of Standby Input, when using the 12 VDC or 24 VDC power supply.
⓮	12VDC	Standby input 12 VDC supply current limited to 20 mA

Item	Name	Description
⓯	24VDC	Standby input 24 VDC supply current limited to 20 mA
⓰	Standby Input +	Positive connection used for Standby Input
⓱	Standby Input –	Negative connection used for Standby Input. Typically connected to ground.

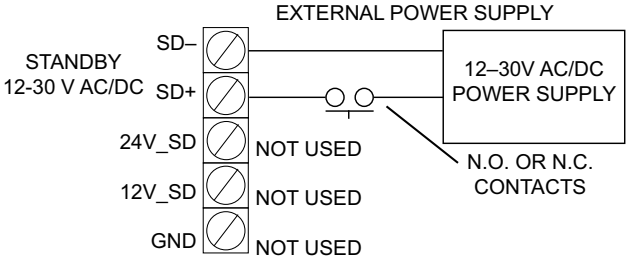
OUTPUT RELAYS: Four output relays connections are available. These relays can be configured, through the user interface, to various alarms detecting normal or abnormal conditions.

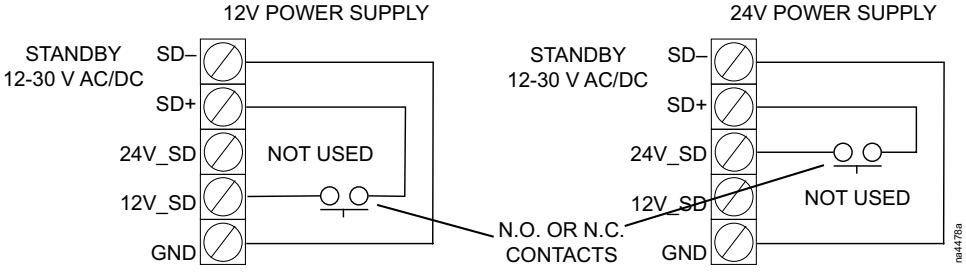
Output relays are Form C type, having a Normally Open (N.O.), Normally Closed (N.C.), and Common (COM) contacts.

NOTE: Output Relays are rated at 24 VDC, 0.6 A maximum.

STANDBY INPUT: Standby input allows the cooling unit to be remotely placed in Standby mode. Standby mode is configured through the user interface and can be set as a normally “active” or “not active” input.

NOTE: Standby Input is rated at 12/24-V AC DC. If using field supplied voltage, the Standby Input consumes 10 mA maximum at 24 VDC.

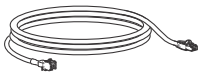




*Normally open contacts shown.

Water Leak Detector—Optional

The leak rope detector (NBES0308) connects to a universal sensor port connection using the supplied cable in the leak detector kit. Route the leak detector cable through one of the low voltage wiring openings.

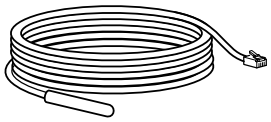


NOTE: The total length of the rope leak sensor can be extended up to 24.4 m (80 ft) using a Rope Leak Sensor Extension kit (NBES0309).

See the documentation supplied with the leak detector kit for more information on the installation.

Remote Temperature sensor

The remote temperature sensor (AP9335T) monitors the room temperature surrounding the cooling unit, ensuring the area is being cooled at the correct temperature.



Location considerations

IMPORTANT: Different locations of the remote temperature sensors can result in different operational behavior of the cooling unit. The optimal position of the temperature sensors will vary from installation to installation, but should be located in an active rack air flow stream to allow accurate readings. It is not recommended to place remote sensors in front of areas with little or no active rack air flow (e.g., in front of blanking panels).

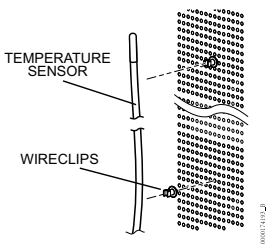
Servers most likely to have insufficient cooling air or inadequately cooled air due to the recirculation of hot air from the hot aisle include the following:

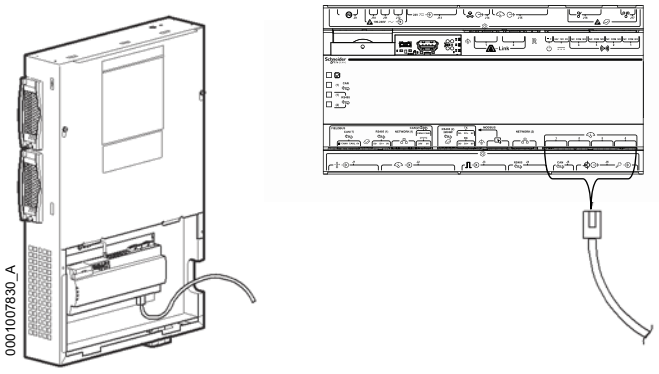
- Servers positioned at the top of a rack
- Servers positioned at any height in the last rack at an open end of a row
- Servers positioned behind flow-impairing obstacles such as building elements
- Servers positioned in a bank of high-density racks
- Servers positioned next to racks with Air Removal Units (ARU)
- Servers positioned very far from or very close to the cooling unit

Installation

NOTE: For proper operation, the temperature sensor must be installed as shown below.

- Route the rack temperature sensor cable through a low voltage wiring opening on the top or bottom of the cooling unit, then through the top or bottom of the adjacent server enclosure.
- Using the provided wire clips, secure the temperature sensor cable to the inside of the front door of an adjacent server enclosure.
- Connect the RJ-45 connector to the controller as shown.



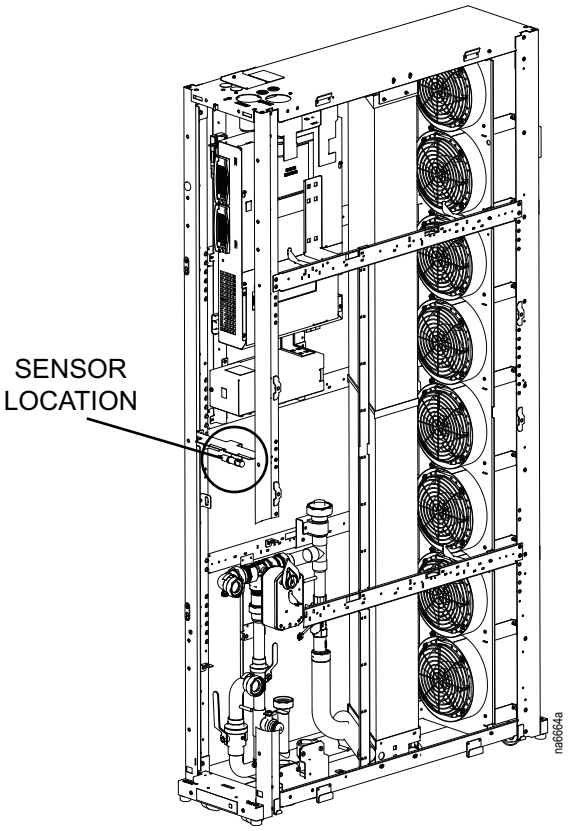


Temperature and Humidity Sensor—ACRC301H

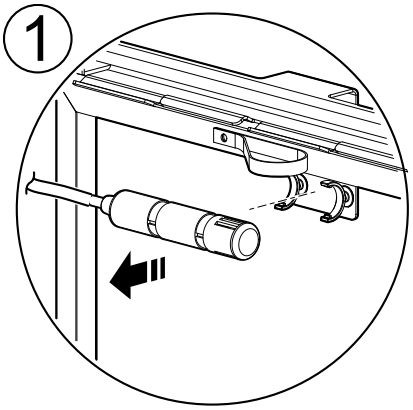
Installation Considerations

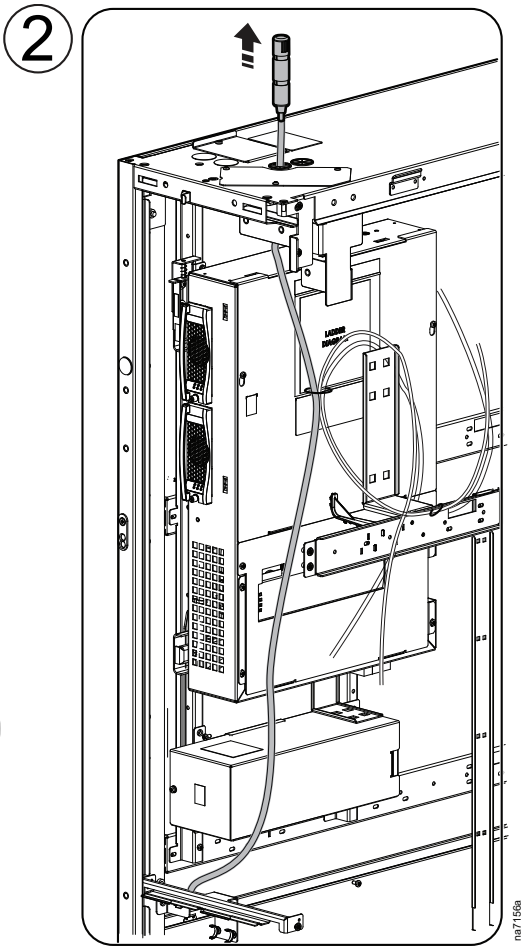
The temperature and humidity sensor located in the middle of the rear of the ACRC301H cooling unit is used to calculate the dew point. It is recommended to relocate the sensor to the front of a rack in the cold aisle (rack air inlet side). It is not recommended to place the sensor in front of areas with little to no active rack air flow.

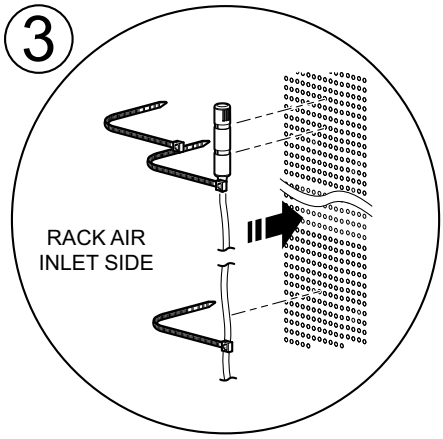
NOTE: The sensor will still operate properly if it is not relocated; however, the readings will not be as accurate.



Installation





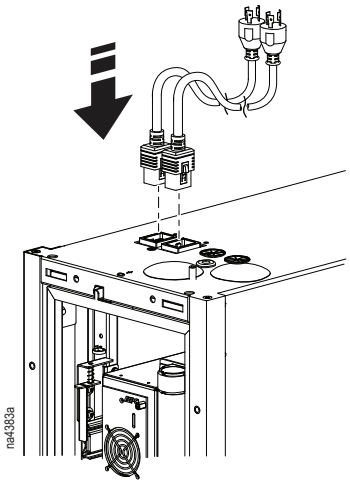


Power Connections

Power cords may be connected at the top of the cooling unit (standard) or at the bottom (optional).

Top Configuration—ACRC301S

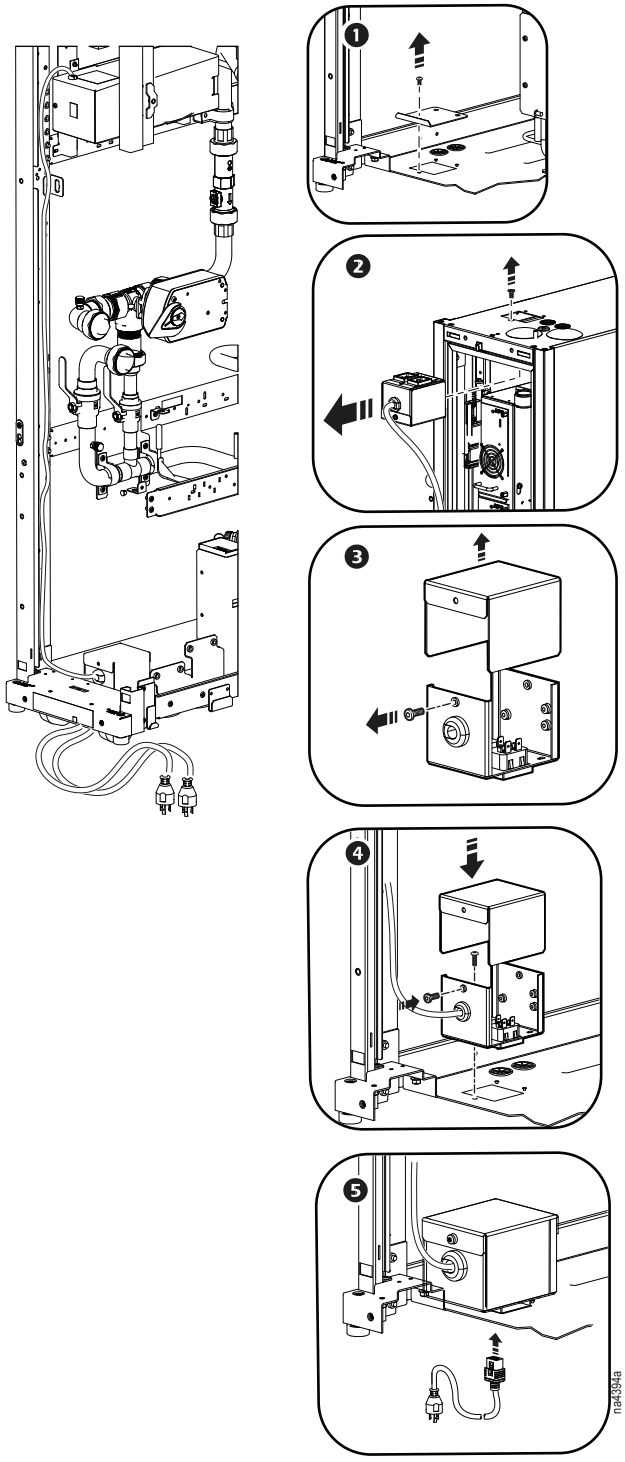
Plug the power cords (supplied) into the electrical outlet located at the top of the cooling unit.



Bottom Configuration (Optional)—ACRC301S

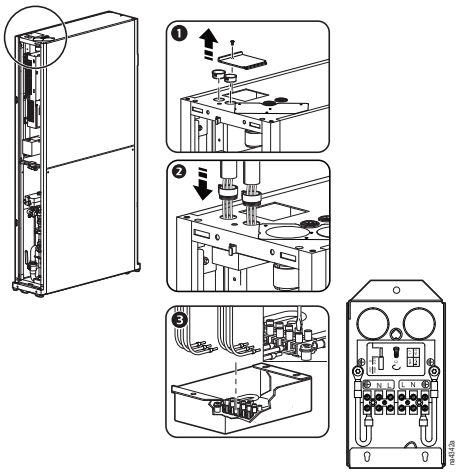
1. Remove the cover plate located at the bottom of the cooling unit. The screw attaching the cover plate will be used to attach the junction box to the bottom of the unit.
2. Remove the junction box from the top of the unit.
3. Remove the cover from the junction box.
4. Relocate the junction box from the top of the cooling unit to the bottom and replace the junction box cover.
5. Plug the power cords (supplied) into the electrical outlets.

   NOTE: Left outlet is secondary power supply; right outlet is the primary power supply (when viewing from the rear of the unit).
6. Place the cover plate on the top of the unit.



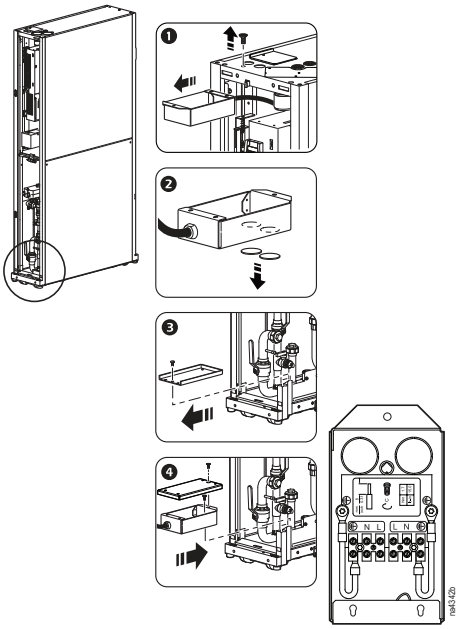
Top Configuration—ACRC301H

1. Remove the cover plate and hole plugs from the top of the unit.
2. Route the power connections through the holes in the top of the unit.
3. Wire connections to terminal blocks.
4. Reinstall the cover plate.



Bottom Configuration—ACRC301H

1. Remove the screw that fastens the box to the top of the unit. Cut and remove wire ties as needed to allow the power cord to be repositioned for bottom power feed connection.
2. Remove the plugs from the holes in the bottom of the junction connection box.
3. Remove the cover plate located at the bottom of the cooling unit. The screw attaching the cover plate will be used to attach the junction box to the bottom of the unit.
4. Attach the junction box to the bottom of the unit. Attach the cover plate to the top of the junction box.
5. Route the power connections through the bottom of the unit.
6. Wire connections to terminal blocks.
7. Reinstall the cover plate.



Primary and Secondary Power Feeds

The equipment can be supplied through two separate power feeds: the primary feed or the secondary feed. Use the display interface to configure the unit to receive power through the primary feed, the secondary feed, or both. The equipment can be supplied with power through the primary feed regardless of whether the secondary feed is powered or not. If power is removed from the primary feed, the secondary feed takes over and supplies power to the equipment (if a secondary feed is connected).

Connect the primary feed and secondary feed input cables to individual, breaker-controlled branch circuits or to PDUs backed by separate Uninterruptible Power Supplies (UPS).

NOTE: The primary and secondary feeds must not use the same branch circuit, PDU, or UPS.



For more information on configuring power feed inputs, see *InRow Chilled Water Air Conditioners Operation and Maintenance Manual*.

First Aid

IMPORTANT: The following measures can be carried out by trained and qualified personnel only.

First Aid Measures in Case of Electrical Shock

In case of electrical shock, follow this procedure:

1. Turn off the electricity source, if possible. If not, remove the same using a dry, non-conducting object made of paperboard, plastic or wood
2. In case of electrical burn, cover any burned areas with a sterile gauze bandage, if available, or a clean cloth. Don't use a blanket or towel, because loose fibers may stick to the burns
3. If necessary, perform mouth-to-mouth resuscitation when the person shows no signs of circulation, such as breathing, coughing or movement

In case of persisting irritations or symptoms, contact the emergency aid.

First Aid Measures in Case of Burn

In case of burn, follow this procedure:

1. Cool the burn: place the burned area under running cool (not cold) water for at least five minutes or apply a clean wet compress to reduce pain and swelling
2. Remove rings or other tight items from the burned area: do this quickly and gently before the area swells
3. Do not break blisters as they prevent infection. If a blister breaks, clean the area with water and apply an antibiotic ointment
4. When a burn is completely cooled, apply a lotion to relieve the area and prevent it from drying
5. Loosely wrap a sterile gauze bandage around the burn
6. If needed, take a pain reliever making sure to carefully read the related information leaflet

In case of severe burn, contact the emergency aid or go to the nearest hospital.

First Aid Measures in Case of Cut from Sharp Edges

In case of deep cuts, follow this procedure:

1. Have the injured person lie down and elevate the site of bleeding. Do not breathe on an open wound. Any objects in the wound should not be removed.
2. Remove or cut clothing around the wound. Gently remove rings or other tight items from the injured area, also to allow blood circulation in case of swelling
3. Apply direct pressure and elevate the area for fifteen minutes. If blood soaks through the cloth, apply another one without lifting the first. If there is an object that could not be removed, apply pressure around the object and not directly over it and then seek medical attention
4. Once the bleeding has stopped, clean the wound to reduce the chance of infection: wash the wound for five minutes with cool water
5. Apply a bandage to protect the cut from dirt and prevent infection. Moisture-enhancing bandages are usually available in first-aid kits

If an infection occurs under the bandage, contact a doctor.

If the cut does not stop bleeding, contact the emergency aid.

First Aid Measures in Case of Fall

In case of fall from considerable heights, do not move the person to avoid further injury.

Call for medical help and, if necessary:

1. Stop any bleeding by applying pressure to the wound with a clean cloth
2. Immobilize the injured area. Do not try and realign any bone sticking out
3. Apply ice packs to limit swelling and help relieve the pain. Do not apply ice directly to the skin

Worldwide Customer Support

Customer support for this product is available at no charge in any of the following ways:

- Visit the Schneider Electric Web site to access documents in the Schneider Electric Knowledge Base and to submit customer support requests.
  - www.se.com (Corporate Headquarters)

    Connect to localized Schneider Electric websites for specific countries, each of which provides customer support information.
  - www.se.com/support/

    Get global support by searching the Schneider Electric Knowledge Base and using esupport.
- Contact the Schneider Electric Customer Support Center by telephone or e-mail.

  Go to **www.se.com > Support > Contact Support** to find contact information for country-specific centers.

For information on how to obtain local customer support, contact the representative or other distributors from whom you purchased your product.

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